

By the Winner
of the
2015 Nobel Prize

THE GREAT ESCAPE

health, wealth,
and the origins
of inequality

ANGUS
DEATON

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Health, Wealth, and the Origins of Inequality

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In memory of Leslie Harold Deaton

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Preface

THE GREAT ESCAPE is a movie about men escaping from a prisoner-of-war camp in World War II. The Great Escape of this book is the story of mankind's escaping from deprivation and early death, of how people have managed to make their lives better, and led the way for others to follow.

One of those lives was my father's. Leslie Harold Deaton was born in 1918 in a tough coal-mining village called Thurcroft in the South Yorkshire coalfield. His grandparents Alice and Thomas had given up agricultural labor in the hope of doing better in the new mine. Their eldest son, my grandfather Harold, fought in World War I, returned to the "pit," and eventually became a supervisor. For my father, it was difficult to become educated in Thurcroft between the wars because only a few children were allowed to go to high school. Leslie took odd jobs at the pit; like the other boys, his ambition was that, one day, he would get the chance to work at the face. He never made it; he was drafted into the army in 1939 and sent to France as part of the ill-fated British Expeditionary Force. After that debacle, he was sent to Scotland to be trained to be a commando; there he met my mother and was "fortunate" enough to be invalided out of the army with tuberculosis and sent to a sanitarium. Fortunate because the commando raid on Norway was a failure, and he would almost certainly have died. He was demobilized in 1942 and married my mother, Lily Wood, the daughter of a carpenter in the town of Galashiels in the south of Scotland.

Although deprived of a high school education in Yorkshire, Leslie had gone to night school to learn surveying skills that were useful in mining, and in 1942, with labor in short supply, those skills made him an attractive hire as an office boy in a firm of civil engineers in Edinburgh. He determined to become a civil engineer himself, and, starting from a base of almost nothing, he put in a decade of hard work and finally qualified. The courses were a great struggle, especially mathematics and physics; the night school he attended, now the Heriot-Watt University in

Edinburgh, recently sent me his examination results, and struggle he certainly did. He took a job as a water supply engineer in the Borders of Scotland and bought the cottage where my mother's grandmother had lived, and where in earlier days Sir Walter Scott was said to have been an occasional visitor. For me, moving from Edinburgh—with its grime, soot, and miserable weather—to a country village—with its woods, hills, and trout streams and, in the summer of 1955, endless sunshine—was a great escape of its own.

In classic manner, my father then set about making sure that I could do better than he had done. Somehow, he managed to persuade my local schoolteachers to coach me outside class for the scholarship examination at a prestigious Edinburgh public (i.e., private) school, where I was one of the two kids in my year that got a free ride; the annual fees were more than my father's salary. I eventually went to Cambridge as a mathematics student and in time became an economics professor, first in Britain and then at Princeton. My sister went to university in Scotland and became a schoolteacher. Of my dozen cousins, we were the only ones to go to university and, of course, none of the previous generation had the chance to do so. Leslie's two grandchildren live in the United States. My daughter is a partner in a successful firm of financial planners in Chicago and my son is a partner in a successful hedge fund in New York. Both received a rich and varied education at Princeton University—vastly superior in its depth, range of opportunities, and quality of teaching to my own dry and narrow experience as an undergraduate in Cambridge. Both of them have a standard of living beyond anything that Leslie could have imagined—though he lived long enough to see a good deal of it, and to be pleased by it. His great-grandchildren live in a world of wealth and opportunity that would have been a far-fetched fantasy in the Yorkshire coalfield.

My father's escape from Thurcroft is an example of what this book is about. He was not born into abject poverty, though it would seem so by today's standards, but he ended his life in comparative affluence. I do not have numbers for the Yorkshire mining villages, but for every thousand children born in England in 1918, more than a hundred did not live to see their fifth

birthday, and the risks would likely have been higher in Thurcroft. Today, children in sub-Saharan Africa are more likely to survive to age 5 than were English children born in 1918. Leslie and his parents survived the great influenza pandemic of 1918–19, though his father died young, killed by a runaway wagon in the pit. My mother’s father died young too, from an infection following an appendectomy. Yet Leslie, in spite of his youthful encounter with tuberculosis—the Captain of Death—lived into his 90th year. His great-grandchildren have a good chance of making it to 100.

Living standards are vastly higher today than a century ago, and more people escape death in childhood and live long enough to experience that prosperity. Almost a century after my father was born, only five out of a thousand British children don’t make it through the first five years, and even if the figure is a little higher in what is left of the Yorkshire coalfield—the Thurcroft pit closed in 1991—it is only a tiny fraction of what it was in 1918. The chance to be educated, so difficult for my father, is taken for granted. Even in my cohort, fewer than one in ten British kids went to college, while today the majority has some form of tertiary education.

My father’s escape, and the future he built for his children and grandchildren, is not an unusual story. Yet it is far from universal. Very few of Leslie’s cohort in Thurcroft ever obtained a professional qualification. My mother’s sisters did not do so, nor did their spouses. Her brother and his family migrated to Australia in the 1960s when their ability to cobble together a bare living from multiple jobs collapsed with the closure of the railway line through the Scottish Borders. My children are financially successful and secure, but they (and we) are extraordinarily fortunate; the children of many well-educated and financially successful people are struggling to do as well as their parents did. For many of our friends, the future of their children and the education of their grandchildren is a constant source of worry.

This is the other side of the story. Even though my father and his family were living longer and prospering in a population that was living longer and prospering on average, not everyone was as motivated or dedicated as my father, nor was everyone as lucky. No one worked harder than my

father, but his luck was important too—the luck not to be among those who died as children, the luck to be rescued from the pit by the war, the luck not to be on the wrong commando raid, the luck not to die from tuberculosis, and the luck to get a job in an easy labor market. Escapes leave people behind, and luck favors some and not others; it makes opportunities, but not everyone is equally equipped or determined to seize them. So the tale of progress is also the tale of inequality. This is especially true today, when the tide of prosperity in the United States is the opposite of equally spread. A few are doing incredibly well. Many are struggling. In the world as a whole, we see the same patterns of progress—of escapes for some, and of others left behind in awful poverty, deprivation, sickness, and death.

This book is about the endless dance between progress and inequality, about how progress creates inequality, and how inequality can sometimes be helpful—showing others the way, or providing incentives for catching up—and sometimes unhelpful—when those who have escaped protect their positions by destroying the escape routes behind them. This is a story that has been told many times, but I want to tell it in a new way.

It is easy to think of the escape from poverty as being about money—about having more and not having to live with the gnawing anxiety of not knowing whether there will be enough tomorrow, dreading that some emergency will come for which there will not be enough funds and which will sink you and your family. Money is indeed a central part of the story. But just as important, or even more so, are better health and the improved likelihood of living long enough to have the chance to prosper. Parents living with the constant fear and frequent reality that their children will die, or mothers giving birth to ten children so that five may survive to adulthood, reflect grisly deprivations that compound the worries about money that haunt many of the same people. Throughout history, and around the world today, the sickness and death of children, endless recurring morbidity for adults, and grinding poverty are partners that often visit the same families, and they do so over and over again.

Many books tell the story of wealth, and many others are about inequality. There are also many books that tell the story of health, and of how health and wealth go hand in hand, with inequalities in health mirroring inequalities in wealth. Here I tell both stories at once, taking the chance that professional demographers and historians will allow an economist to trespass into their lands. But the story of human wellbeing, of what makes life worth living, is not well served by looking at only a part of what is important. The Great Escape does not respect the boundaries of academic disciplines.

I have accumulated many intellectual debts throughout my life as an economist. Richard Stone was perhaps the most profound influence; from him I learned about measurement—how little we can say without it and how important it is to get it right. From Amartya Sen I learned how to think about what makes life worth living and about how we must study wellbeing as a whole, not just parts of it. The measurement of wellbeing is the heart of this book.

My friends, colleagues, and students have been extraordinarily generous in reading drafts of all or parts of this book. It is immeasurably better for their thoughtful and insightful reactions. I am especially grateful to those who disagree with me, yet who took the time not only to criticize and persuade but also to praise and agree when they could. I am grateful to Tony Atkinson, Adam Deaton, Jean Drèze, Bill Easterly, Jeff Hammer, John Hammock, David Johnston, Scott Kostyshak, Ilyana Kuziemko, David Lam, Branko Milanovic, Franco Peracchi, Thomas Pogge, Leandro Prados de las Escosura, Sam Preston, Max Roser, Sam Schulhofer-Wohl, Alessandro Tarozzi, Nicolas van de Walle, and Leif Wenar. My editor at Princeton University Press, Seth Ditchik, helped me to get started and provided help and good advice all along the way.

Princeton University has provided me with an unequalled academic environment for more than three decades. The National Institute of Aging and the National Bureau of Economic Research have helped fund my work on health and wellbeing, and the results from that research have influenced this book. I have frequently worked with the World Bank; the Bank is constantly

faced with urgent, practical problems and has helped teach me which issues matter and which do not. In recent years, I have been a consultant to the Gallup Organization; they have pioneered the global investigation of wellbeing, and some of the information that they have collected appears in the early part of the book. I am grateful to all of them.

Last, and most, Anne Case read every word soon after it was written, and sometimes many times afterwards. She is responsible for innumerable improvements throughout, and without her endless encouragement and support this book would not exist.

INTRODUCTION

What This Book Is About

LIFE IS BETTER NOW than at almost any time in history. More people are richer and fewer people live in dire poverty. Lives are longer and parents no longer routinely watch a quarter of their children die. Yet millions still experience the horrors of destitution and of premature death. The world is hugely unequal.

Inequality is often a *consequence* of progress. Not everyone gets rich at the same time, and not everyone gets immediate access to the latest life-saving measures, whether access to clean water, to vaccines, or to new drugs for preventing heart disease. Inequalities in turn affect progress. This can be good; Indian children see what education can do and go to school too. It can be bad if the winners try to stop others from following them, pulling up the ladders behind them. The newly rich may use their wealth to influence politicians to restrict public education or health care that they themselves do not need.

This book tells stories of how things got better, how and why progress happened, and the subsequent interplay of progress and inequality.

The Great Escape: The Movie

The Great Escape, a famous movie about prisoners of war in World War II, is based on the exploits of Roger Bushell (in the film, Roger Bartlett, played by Richard Attenborough), a South African in the Royal Air Force who was shot down behind German lines, and who repeatedly escaped and was repeatedly recaptured.¹ In his third attempt, as depicted in the film—the Great Escape—250 prisoners escaped with him through tunnels dug from Stalag Luft III. The movie tells the story of how the escape was planned; the ingenuity that went into constructing three tunnels,

Tom, Dick, and Harry; and the improvisation and technical skills that went into making civilian clothes and forging papers, all under the eyes of the watchful guards. All but three of the POWs were eventually recaptured, and Bushell himself was executed on direct orders from Hitler. Yet the emphasis of the movie is not on the limited success of this particular escape, but on man's unquenchable desire for freedom, even under impossibly difficult circumstances.

In this book, when I speak of freedom, it is the freedom to live a good life and to do the things that make life worth living. The absence of freedom is poverty, deprivation, and poor health—long the lot of much of humanity, and still the fate of an outrageously high proportion of the world today. I will tell stories of repeated escapes from this kind of prison, how and why they came about, and what happened afterwards. It is a story of material and physiological progress, of people becoming richer and healthier, of escapes from poverty.

A phrase in my subtitle, “the origins of inequality,” comes from thinking about the POWs who did not escape. All of the POWs could have stayed where they were, but instead a few escaped, some died, some were returned to the camp, and some never left. This is in the nature of most “great escapes”: not everyone can make it, a fact that in no way makes the escape less desirable or less admirable. Yet when we think about the consequences of the escape, we need to think not just about those who were the heroes of the movie, but also about those who were left behind in Stalag Luft III and other camps. Why should we care about them? The movie certainly did not; they are not the heroes and are incidental to the story. There is no movie *The Great Left Behind*.

Yet we *should* think about them. After all, the number of POWs in German camps who did not escape was far greater than the few who did. Perhaps they were actually harmed by the escape, if they were punished or if their privileges were withdrawn. One can imagine that the guards made it even harder to escape than before. Did the escape of their fellow POWs inspire those still in the camps to escape too? They certainly could have learned from the escape techniques developed by

the Great Escapees, and they might have been able to avoid their mistakes. Or were they discouraged by the difficulties or by the very limited success of the Great Escape itself? Or perhaps, jealous of the escapees and pessimistic about their own chances, they became unhappy and depressed, making camp conditions even worse.

As with all good movies, there are other interpretations. The success and exhilaration of the escape are all but extinguished by the end of the film; for most of the escapees, their freedom is only temporary. Humanity's escape from death and deprivation began around 250 years ago, and it goes on to this day. Yet there is nothing to say that it must continue forever, and many threats—climate change, political failures, epidemics, and wars—could bring it to an end. Indeed, there were many pre-modern escapes in which rising living standards were choked off by precisely such forces. We can and should celebrate the successes, but there is no basis for a thoughtless triumphalism.

Economic Growth and the Origins of Inequality

Many of the great episodes of human progress, including those that are usually described as being entirely good, have left behind them a legacy of inequality. The Industrial Revolution, beginning in Britain in the eighteenth and nineteenth centuries, initiated the economic growth that has been responsible for hundreds of millions of people escaping from material deprivation. The other side of the same Industrial Revolution is what historians call the “Great Divergence,” when Britain, followed a little later by northwestern Europe and North America, pulled away from the rest of the world, creating the enormous gulf between the West and the rest that has not closed to this day.² Today's global inequality was, to a large extent, created by the success of modern economic growth.

We should not think that, prior to the Industrial Revolution, the rest of the world had always been backward and desperately poor. Decades before Columbus, China was advanced and

rich enough to send a fleet of enormous ships under Admiral Zheng He—aircraft carriers relative to Columbus’s rowboats—to explore the Indian Ocean.³ Three hundred years before even that, the city of Kaifeng was a smoke-filled metropolis of a million souls whose belching mills would not have been out of place in Lancashire eight hundred years later. Printers produced millions of books that were cheap enough to be read by people of even modest means.⁴ Yet those eras, in China and elsewhere, were not sustained, let alone taken as starting points for ever-increasing prosperity. In 1127, Kaifeng fell to an invasion of tribes from Manchuria who had been rashly employed to help it wage war; if you enlist dangerous allies, you had better make sure they are well paid.⁵ Economic growth in Asia kept starting and kept being choked off, by rapacious rulers, by wars, or by both.⁶ It is only in the last two hundred and fifty years that long-term and continuing economic growth in some parts of the world—but not in others—has led to persistent gaps between countries. Economic growth has been the engine of international income inequality.

The Industrial Revolution and the Great Divergence are among the more benign escapes in history. There are many occasions when progress in one country was at the *expense* of another. The Age of Empire in the sixteenth and seventeenth centuries, which preceded the Industrial Revolution and helped cause it, benefited many in England and Holland, the two countries that did best in the scramble. By 1750, laborers in London and Amsterdam had seen their incomes grow relative to laborers in Delhi, Beijing, Valencia, and Florence; English workers could even afford a few luxuries, such as sugar and tea.⁷ Yet those who were conquered and plundered in Asia, Latin America, and the Caribbean were not only harmed at the time but in many cases saddled with economic and political institutions that condemned them to centuries of continuing poverty and inequality.⁸

Today’s globalization, like earlier globalizations, has seen growing prosperity alongside growing inequality. Countries that were poor not long ago, like China, India, Korea, and Taiwan, have taken advantage of globalization and grown rapidly, much faster than have today’s rich

countries. At the same time, they have moved away from still poorer countries, many of them in Africa, creating new inequalities. As some escape, some are left behind. Globalization and new ways of doing things have led to continuing increases in prosperity in rich countries, though the rates of growth have been slower—not only than in the fast-growing poor countries, but also than they used to be in the rich countries themselves. As growth has slowed, gaps between people have widened *within* most countries. A lucky few have made fabulous fortunes and live in a style that would have impressed the greatest kings and emperors of centuries past. Yet the majority of people have seen less improvement in their material prosperity, and in some countries—the United States among them—people in the middle of the income distribution are no better off than were their parents. They remain, of course, many times better off than still earlier generations; it is not that the escape never happened. Yet many today have good reasons to worry whether their children and grandchildren will look back to the present not as a time of relative scarcity but as a long-lost golden age.

When inequality is the handmaiden of progress, we make a serious mistake if we look only at average progress or, worse still, at progress only among the successes. The Industrial Revolution used to be told as a story of what happened in the leading countries, ignoring the rest of the world—as if nothing was happening there, or as if nothing had *ever* happened there. This not only slighted the majority of mankind but also ignored the unwilling contributions of those who were harmed or, at best, just left behind. We cannot describe the “discovery” of the New World by looking only at its effects on the Old. Within countries, the average rate of progress, such as the rate of growth of national income, cannot tell us whether growth is widely shared—as it was in the United States for a quarter of a century after World War II—or is accruing to a small group of very wealthy people—as has been the case more recently.

I tell the story of material progress, but that story is one of *both* growth and inequality.

Not Just Income, but Health Too

Progress in health has been as impressive as progress in wealth. In the past century, life expectancy in the rich countries increased by thirty years, and it continues to increase today by two or three years every ten years. Children who would have died before their fifth birthdays now live into old age, and middle-aged adults who once would have died of heart disease now live to see their grandchildren grow up and go to college. Of all the things that make life worth living, extra years of life are surely among the most precious.

Here too progress has opened up inequalities. The knowledge that cigarette smoking kills has saved millions of lives in the past fifty years, yet it was educated, richer professionals who were the first to quit, opening up a health gap between rich and poor. That germs caused disease was new knowledge around 1900, and professionals and educated people were the first to put that knowledge into practice. We have known for the best part of a century how to use vaccines and antibiotics to stop children from dying, yet around two million children still die every year from vaccine-preventable disease. Rich people are treated in world-class modern medical facilities in São Paulo or Delhi while, a mile or two away, poor children are dying of malnutrition and easily preventable disease. The explanation for why progress should be so uneven differs from case to case; the reason why poor people are more likely to smoke is not the same as the reason why so many poor children are not vaccinated. These accounts are to come, but for now the point is simply that health progress creates gaps in health just as material progress creates gaps in living standards.

These “health inequalities” are one of the great injustices of the world today. When new inventions or new knowledge comes along, *someone* has to be the first to benefit, and the inequalities that come with waiting for a while are a reasonable price to pay. It would be absurd to wish that knowledge about the health effects of smoking had been suppressed so as to prevent new health inequalities. Yet poor people are still more likely to smoke, and the children who are dying today in Africa would not have died in France or the United States even sixty years ago. Why do

these inequalities persist, and what can be done about them?

This book is mostly about two topics: material living standards and health. They are not the only things that matter for a good life, but they are important in and of themselves. Looking at health and income together allows us to avoid a mistake that is too common today, when knowledge is specialized and each specialty has its own parochial view of human wellbeing. Economists focus on income, public health scholars focus on mortality and morbidity, and demographers focus on births, deaths, and the size of populations. All of these factors contribute to wellbeing, but none of them *is* wellbeing. The statement is obvious enough, but the problems that arise from it are not so obvious.

Economists—my own tribe—think that people are better off if they have more money—which is fine as far as it goes. So if a few people get a lot more money and most people get little or nothing, but do not lose out, economists will usually argue that the world is a better place. And indeed there is enormous appeal to the idea that, as long as no one gets hurt, better off is better; it is called the Pareto criterion. Yet this idea is completely undermined if wellbeing is defined too narrowly; people have to be better off, or no worse off, in *wellbeing*, not just in material living standards. If those who get rich get favorable political treatment, or undermine the public health or public education systems, so that those who do less well lose out in politics, health, or education, then those who do less well may have gained money but they are *not* better off. One cannot assess society, or justice, using living standards alone. Yet economists routinely and incorrectly apply the Pareto argument to income, ignoring other aspects of wellbeing.

Of course, it is also a mistake to look at health, or at any one *component* of wellbeing, by itself. It is a good thing to improve health services, and to make sure that those who are in medical need are looked after. But we cannot set health priorities without attention to their cost. Nor should we use longevity as a measure of social progress; life in a longer-lived country is better, but not if the country is a totalitarian dictatorship.

Wellbeing cannot be judged by its average without looking at inequality, and wellbeing cannot be judged by one or more of its parts without looking at the whole. If this book were much longer, and its author knew much more, I would write about other aspects of well-being, including freedom, education, autonomy, dignity, and the ability to participate in society. But even thinking about health and income in the same book will free us from the mistakes that come from looking at one or the other alone.

How Does Progress Come About?

There is little doubt that our ancestors would have liked to have what we have now, could they have imagined our world. And there is no reason to think that parents ever become inured to watching their children die; if you doubt me (and it is only one account among many), read Janet Browne's description of the tortures suffered by Charles Darwin when his first two children died.⁹ The desire to escape is always there. Yet the desire is not always fulfilled. New knowledge, new inventions, and new ways of doing things are the keys to progress. Sometimes inspiration comes from lone inventors who dream up something quite different from what has gone before. More often, new ways of doing things are by-products of something else; for example, reading spread when Protestants were required to read the Bible for themselves. More often still, the social and economic environment creates innovations in response to need. Wages were high in Britain after its success in the Age of Empire, and those high wages, together with plentiful coal, provided incentives for inventors and manufacturers to come up with the inventions that powered the Industrial Revolution.¹⁰ The British Enlightenment, with its relentless search for self-improvement, provided fertile intellectual soil in which those inventions were more likely to come about.¹¹ The cholera epidemics of the nineteenth century were an impetus for crucial discoveries about the germ theory of disease. And the well-funded medical research arising from the HIV/AIDS pandemic of today uncovered the virus and developed medicines that, while not

curing the disease, greatly extend the lives of those who are infected. Yet there are also cases in which inspiration never came, in which needs and incentives failed to produce a magic solution, or even a mundane one. Malaria has afflicted human beings for tens of thousands of years, perhaps even for all of human history, and we still have no comprehensive way of preventing or treating it. Necessity may be the mother of invention, but there is nothing that guarantees a successful pregnancy.

Inequality also influences the process of invention, sometimes for good and sometimes for ill. The sufferings of the deprived are a force for finding new ways to close the gaps, if only because the fact that some are not deprived demonstrates that the deprivation need not exist. A good example is the discovery of oral rehydration therapy in the refugee camps of Bangladesh in the 1970s; millions of children suffering from diarrhea have been saved from dehydration and possible death by a cheap and easily made remedy. But it works the other way too. Powerful interests have much to lose from new inventions and new ways of doing things. Economists think of eras of innovation as powering up waves of “creative destruction.” New methods sweep away old methods, destroying the lives and livelihoods of those who were dependent on the old order. Globalization today has hurt many such groups; importing cheaper goods from abroad is like a new way of making them, and woe betide those who earned their livings making such goods at home. Some of those who would lose out, or who fear that they might be hurt, are politically powerful and can outlaw or slow down the new ideas. The emperors of China, worried about threats to their power from merchants, banned oceangoing voyages in 1430, so that Admiral Zheng He’s explorations were an end, not a beginning.¹² Similarly, Francis I, Emperor of Austria, banned railways because of their potential to bring about revolution and threaten his power.¹³

Why Does Inequality Matter?

Inequality can spur progress or it can inhibit progress. But does it matter in and of itself?

There is no general agreement on this: the philosopher and economist Amartya Sen argues that even among the many who believe in some form of equality, there are very different views about *what* it is that ought to be made equal.¹⁴ Some economists and philosophers argue that inequalities of income are unjust, unless they are necessary for some greater end. For example, if a government were to guarantee the same income for all of its citizens, people might decide to work a lot less so that even the very poorest might be worse off than in a world in which some inequality is allowed. Others emphasize equality of opportunity rather than equality of outcomes, though there are many versions of what equality of opportunity means. Yet others see fairness in terms of proportionality: what each person receives should be proportional to what he or she contributes.¹⁵ On this view of fairness, it is easy to conclude that income *equality* is unfair if it involves redistributing income from rich to poor.

In this book, the arguments I emphasize are those about what inequality does, whether inequality helps or hurts, and whether it matters what kind of inequality we are talking about. Does society benefit from having very rich people when most are not rich? If not, does society benefit from the rules and institutions that allow some to get much richer than the rest? Or do the rich harm everyone else—for example by making it difficult for the nonrich to affect how society is run? Are inequalities in health like inequalities in income, or are they somehow different? Are they always unjust, or can they sometimes serve a higher good?

A Road Map

The aim of the book is to provide an account of wealth and health around the world, focusing on today but also looking back to see how we got to where we are. Chapter 1 is an introductory overview. It gives a snapshot of the world from outer space: a map of where life is good and of where it is not good. It documents a world in which there has been great progress in reducing poverty and lowering the chances of death, but also a world of difference—of huge

inequalities in living standards, in life chances, and in wellbeing.

The three chapters of Part I are about health. They look at how the past has shaped our health today, why the hundreds of thousands of years that people spent as hunter-gatherers are relevant for understanding health today, and why the mortality revolution that began in the eighteenth century set patterns that are echoed in contemporary health advances. The move to agriculture, seven to ten thousand years ago, made it possible to grow more food, but it also brought new diseases and new inequalities as hierarchic states replaced egalitarian bands of hunter-gatherers. In England of the eighteenth century, globalization brought new medicines and new treatments that saved many lives—but mostly the lives of those who could afford them. While the new methods eventually lowered death rates for everyone, it was the aristocracy whose life chances first pulled away from those of the common people. By the end of the nineteenth century, the development and acceptance of the germ theory of disease had set the stage for another explosion of progress as well as for the opening up of another great chasm—this time between the life chances of those who were born in rich countries and the chances of those who were not.

I tell the story of the fight to save the lives of children in the world that was left behind. This is a story of progress, mostly after World War II—a catch-up that would begin to close the chasm that had begun to open in the eighteenth century. It is a story with many great successes, in which antibiotics, pest control, vaccinations, and clean water saved millions of children, and in which life expectancy sometimes increased at (the apparently impossible rate of) several years each year. The chasm in life expectancy between the poor and rich worlds was narrowed, but not closed. There were also terrible setbacks, including a catastrophic man-made famine in China between 1958 and 1961, and the recent HIV/AIDS epidemic that, for several African countries, wiped out three decades of progress against mortality. Even without those disasters, much remains undone; many countries do not have adequate systems for routine health care, many children still die just because they were born in the “wrong” country, and there remain places—most notably

but not only in India—where half of the children are seriously malnourished.

One of the (good) reasons why the mortality gap between rich and poor has not closed more rapidly is because mortality has been falling in rich countries too, but in a very different way, benefiting children less and adults more. The final installment of the health story is about mortality decline in rich countries, about how and why the gap in life expectancy between men and women has been closing, about the (huge) role played by cigarette smoking, and about why the fight against heart disease has been so much more successful than the fight against cancer. Once again we see progress coupled with growing health inequalities, just as happened in Britain in the late eighteenth century.

The two chapters of Part II are about material living standards. I start with the United States; although America is indeed exceptional and is often extreme, for example in its degree of income inequality, the forces at work apply to other rich countries too. Economic growth brought new prosperity to Americans after World War II, but growth had been slowing, decade by decade, even before the Great Recession. Postwar growth brought marked reductions in poverty, especially among African-Americans and the elderly, and there was little expansion in inequality. Until the early 1970s, the United States was the very model of a modern major economy. Since then, the story has been one of less growth and greater inequality, the latter driven especially by runaway growth in incomes at the very top of the distribution. As always, there is a good side to this inequality: rewards to education, to innovation, and to creativity are higher than they have ever been. But the United States is also a good example of the dark side, of the political and economic threats to wellbeing that come from plutocracy.

I also look at living standards in the world as a whole. Here is the story of perhaps the greatest escape in all of human history, and certainly the most rapid one: the reduction in global poverty since 1980. Much of it was driven by the performance of the two largest countries in the world, China and India, where recent economic growth has transformed the lives of more than a

billion people. That global poverty should have fallen goes against the almost universally accepted doomsday predictions of the 1960s, that the population explosion would doom the world to deprivation and disaster. The world has done much better than the pessimists predicted. Yet a billion or so people *still* live in terrible destitution; while many have escaped, many have been left behind.

Part III consists of a single chapter, an epilogue in which I stop telling stories and argue for what ought to be done—and more importantly for what ought not to be done. I believe that we—meaning those of us who are fortunate enough to have been born in the “right” countries—have a moral obligation to help reduce poverty and ill health in the world. Those who have escaped—or at least have escaped through the struggles of their predecessors—must help those who are still imprisoned. For many people, that moral duty is fulfilled by foreign aid, through the efforts of national governments (most of whom have official aid agencies), through international organizations like the World Bank or the World Health Organization, or through the thousands of nongovernmental aid organizations that operate nationally and internationally. While some of this aid has clearly done good—and I think the case for assistance to fight disease such as HIV/AIDS or smallpox is strong—I have come to believe that most external aid is doing more harm than good. If it is undermining countries’ chance to grow—as I believe it is—there is no argument for continuing it on the grounds that “we must do something.” The something that we should do is to stop.

The Postscript is a coda that returns to the main themes. It asks whether we can expect the real Great Escape—unlike the movie *The Great Escape*—to have a happy ending.

Measuring Progress, Measuring Inequality

Whenever it is possible to do so, I support my arguments with data, and almost always with graphs. Progress cannot be coherently discussed without definitions and supporting evidence.

Indeed, enlightened government is impossible without the collection of data. States have been counting their populations for thousands of years—the Roman census that sent Mary and Joseph to Bethlehem, Joseph’s city of birth, is a famous example. The U.S. Constitution mandates that there be a census of the population every ten years; without it, a fair democracy is not possible. Even earlier, in 1639, the colonists in present-day Massachusetts mandated a complete count of births and deaths; without such vital statistics, public health policy is blind.

Not the least of the health problems faced by the poor countries of the world today is the lack of good information on the numbers of people who die, let alone on what causes their deaths. There is no lack of invented and interpolated numbers from international agencies, but it is not always widely understood that these are not an adequate basis for policy or for thinking about or assessing external aid. The need to do something tends to trump the need to understand what needs to be done. And without data, anyone who does anything is free to claim success. As I go along, I will try to explain the basis for my numbers, where they come from and how credible (or incredible) they are. I will also try to make the case that the missing data are a scandal that is not being adequately addressed.

Unless we understand how the numbers are put together, and what they mean, we run the risk of seeing problems where there are none, of missing urgent and addressable needs, of being outraged by fantasies while overlooking real horrors, and of recommending policies that are fundamentally misconceived.

National Happiness and National Income

Much of this book is about material wellbeing, typically measured by income, the amount of money that people have to spend or to save. Money must always be adjusted by the costs of what people buy, but, that done, it is a reasonable indicator of people’s ability to buy the things on which material wellbeing depends. Yet many argue that too much attention is given to income. A good

life certainly means more than money, but the argument often goes further, to claim that money does nothing to make people's lives better, at least once basic needs have been met.

Some evidence for this argument comes from happiness surveys that show, it is claimed, that money does little or nothing to make people happy except for those in poverty. If this is correct, and if happiness is the right way to measure wellbeing, then much of my argument would be undercut. So it is good to start out by considering how happiness relates to money. The discussion will also give me the chance to introduce and explain a way of drawing graphs that I will use throughout the book.

Surveys often ask people how their lives are going, for example by reporting how satisfied they are with their lives in general. These data are often referred to as measures of “happiness,” though it is easy to think of examples in which unhappy people believe that their lives are going well, or vice versa. Indeed, as we shall see, it is a bad mistake to confuse life satisfaction and happiness; the former is an overall judgment about life that comes from consideration, while the latter is an emotion, a mood, or a feeling, which is part of experiencing life.¹⁶

The Gallup Organization asks people around the world to rate their lives by imagining a “ladder of life” with eleven steps; the bottom step, 0, is “the worst possible life for you” while 10 is “the best possible life for you.” Each respondent is asked to indicate “on which step of the ladder would you say you personally feel you stand at this time?” We can use these data to see how countries do relative to one another and, in particular, whether higher-income countries do better on this measure.

Figure 1 shows the average life evaluation for each country against its national income per head, or more precisely gross domestic product (GDP) per head; it shows the averages for the years 2007 through 2009. Income is measured in U.S. dollars that have been adjusted for differences in price levels between countries; in Chapter 6, I will explain where these numbers come from as well as the considerable reservations that should be attached to them. The circles in the figure have

areas proportional to the populations of each country; the two big countries on the left are China and India, and the big country at the top right is the United States. I have marked a few other countries that are particularly interesting.

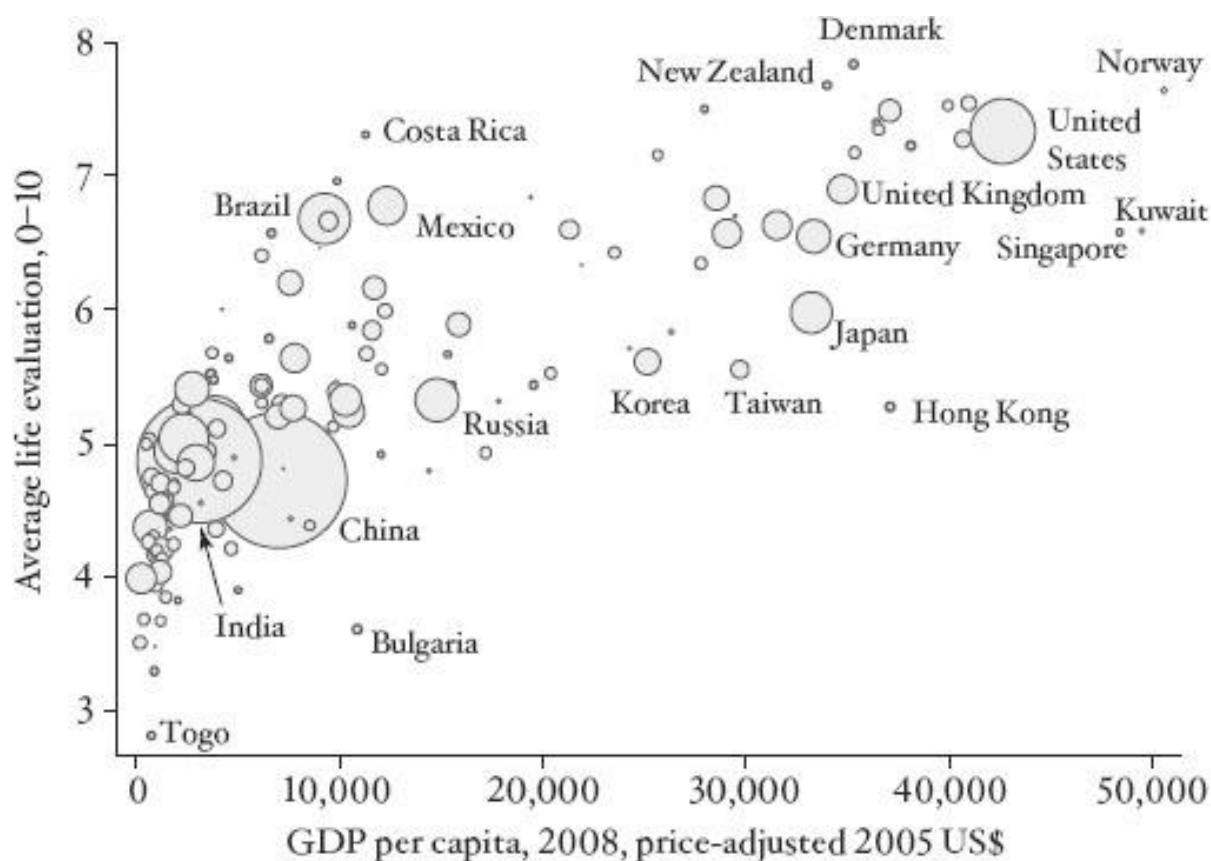


FIGURE 1 Life evaluation and GDP per capita.

We can see at once that the people who live in the really poor countries on the left of the figure are generally very dissatisfied with their lives; not only are they poor in income, but they also rate their lives poorly. At the other end of the world, in the United States and the other rich countries, people have high incomes and evaluate their lives highly. The worst country is Togo—one of the poorest countries in the world, where people have very little freedom of any kind—while the best is Denmark—a rich, free country. The Scandinavian countries regularly

outrank the United States in these comparisons, but the average life evaluation in the United States is still among the best in the world. There are lots of exceptions to the rule of income. East Asian countries and former communist countries tend to have low life evaluations—Bulgaria is the most extreme example—while countries in Latin America tend to do relatively well. Income is certainly not the only thing that matters in people’s evaluations of their lives.

If we look at the bottom left of the picture, where the poor countries are, we see that life evaluation rises with national income quite rapidly. After we pass China and India, traveling from bottom left to top right, the rise in life evaluation with income is a bit less steep, and once we get to Brazil and Mexico the life evaluation scores are close to seven out of ten, only a point or so less than those for the really rich countries at the top right. Income matters more among the very poor than among the very rich. Indeed, it is very tempting to look at the picture and conclude that once GDP per capita reaches around \$10,000 a year, more money does nothing to improve people’s lives, and many have made this claim.¹⁷ Yet this claim is false.

To explain why money matters even among the rich countries, we need to redraw Figure 1 in a somewhat different form. When we think about money, we think in dollar terms, but we also think in percentage terms. On the rare occasions when my Princeton colleagues discuss their salaries with one another, they are likely to report that one got a 3 percent increase, while another got 1 percent. Indeed, the dean is more likely to signal his pleasure or displeasure through the size of the percentage increase than through the size of the increase in dollars. While a 1 percent increase means more dollars to someone who earns \$200,000 a year than a 2 percent increase means to someone earning \$50,000 a year, the latter will (correctly) feel that she has done better in the past year. Percentage changes become the basic unit in these sorts of calculations; 10 percent is the same no matter what the baseline income is.

We can do just this for the data in Figure 1, though the differences between countries are so huge that it makes sense to think, not in terms of percentages, but in terms of the number of times

income is quadrupled. Think of \$250 a year as the base; only Zimbabwe and the Democratic Republic of the Congo (DRC) are at or below \$250. Countries such as Uganda, Tanzania, and Kenya are near \$1,000, four times the base; China and India are another fourfold increase over Tanzania and Kenya, near the marker for sixteen times the base. Mexico and Brazil are four times China and India, and the world's richest countries have incomes that are four times larger still; they are 256 times richer than the world's poorest countries. (In Chapter 6 I shall explain why these numbers should only be taken as rough guides.) Instead of using the dollar value of incomes to compare with life evaluations, we can use this scheme of fourfold comparison, marking off units as 4 times, 16 times, 64 times, and 256 times the base, and this is what is done in Figure 2.

Figure 2 contains *exactly the same data* as Figure 1, but income is now plotted on this 1, 4, 16, 64, and 256 scale. I have, however, marked these five points by their original dollar amounts, \$250 through \$64,000 so that the link with income itself is clear. Moving along the horizontal axis from one tick to the next always represents a fourfold increase in income. More generally, equal distances from left to right represent equal percentage increases in income, not equal dollar amounts, as in Figure 1. A scale with this property is known as a *logarithmic* (or log) scale, and we will see it again.

Although the only change is in the labeling of the horizontal axis, Figure 2 looks completely different from Figure 1. The flattening among the rich countries has vanished, and the countries now lie more or less along a straight line. What this says is that equal *percentage* differences in income produce equal absolute shifts in life evaluation. On average, if we move from one country to another whose per capita income is four times as high, the life evaluation score will move by about one point on a zero to ten scale, and this is true whether we are moving between poor countries or rich ones. And just to remove any misunderstanding: yes, there are lots of exceptions, and lots of countries are higher or lower than we might expect them to be given their national incomes. It is not always true that all rich countries have higher life evaluations than all

the countries that are poorer; China and India are two notable examples. But on average over all the countries, rich or poor, a fourfold difference in incomes comes with a one-point increase in the evaluation of life.

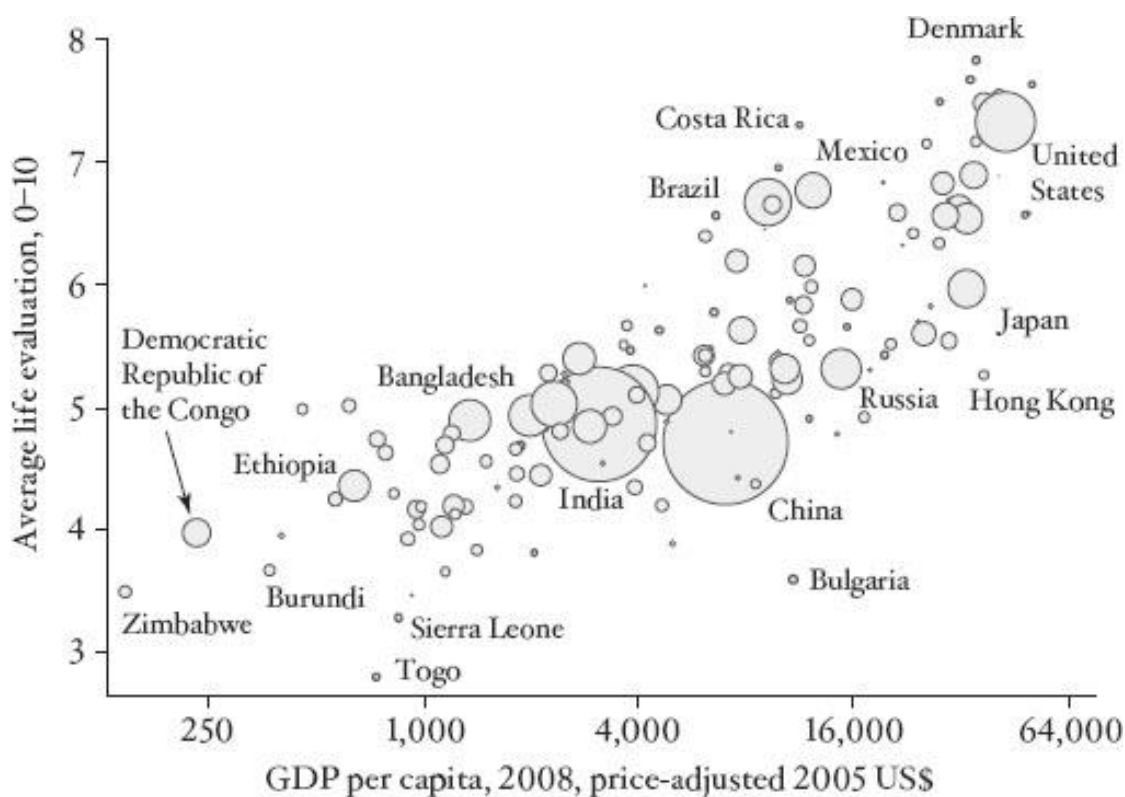


FIGURE 2 Life evaluation and GDP per capita on a log scale.

Is Figure 1 right, or is Figure 2 right? Both are, just as it is true that the professor who got a 2 percent raise on \$50,000 got a raise of \$1,000 while the professor who got a 1 percent raise on \$200,000 got \$2,000. The same percentage increase involves more money if we move from India to the United States than if we move from the DRC to India, even though both involve a fourfold shift. Figure 1 tells us that the same *absolute* increase in dollars means less to the life satisfaction of a rich person than to a poor person, while Figure 2 tells us that the same *percentage* increase makes for the same increase in life satisfaction.

Life evaluation scores capture important aspects of life beyond income, and this has led to arguments that we should downplay the importance of income. This is fine if the implication is to consider other aspects of wellbeing, like health or education or the ability to participate in society. It is not fine if the implication is that income is not worth anything, or that income adds nothing to life for those of us who live in countries richer than Mexico. It is even less fine if the argument is that we should focus on life evaluations and ignore everything else. Life evaluation measures are far from perfect. People are not always sure what the questions mean, or how they are expected to answer, and international comparisons can be compromised by national differences in reporting styles. In many places, “mustn’t grumble” or “not so bad” is about as good as anyone would ever claim, but people in other cultures are more exuberant about their feelings and less reticent about their successes. So Figure 2 is important because it shows that focusing on income is not seriously misleading. Richer countries have higher life evaluations, even among the world’s richest countries.

I shall return to measures of happiness and life satisfaction in the next chapter, but my main purpose there is to look more widely at the wellbeing of the world today—at those who have made the Great Escape, and then some, as well as those who are still waiting.

ONE

The Wellbeing of the World

THE GREATEST ESCAPE in human history is the escape from poverty and death. For thousands of years, those who were lucky enough to escape death in childhood faced years of grinding poverty. Building on the Enlightenment, the Industrial Revolution, and the germ theory of disease, living standards have increased by many times, life spans have more than doubled, and people live fuller and better lives than ever before. The process is still going on. My father lived twice as long as either of my grandfathers; his real income as a civil engineer was many times the income of his father, who was a coal miner; and my education and income as a professor greatly exceed his education and his income. Mortality rates of children and of adults continue to fall throughout the world. But the escape is far from complete. A billion people suffer living standards, schooling, and life spans that are little better than those of their (or our) forebears. The Great Escape has made a world of difference to those of us who are richer, healthier, taller, bigger, and better educated than our grandparents and their grandparents. It has also made a world of difference in another, less positive sense: because much of the world's population was left behind, the world is immeasurably more unequal than it was three hundred years ago.

This book tells the story of the Great Escape, of the benefits to mankind that it brought, and how it was responsible for today's unequal world. It also explains what we need to do—or not to do—to help those who are still trapped in deprivation.

I use the term *wellbeing* to refer to all of the things that are good for a person, that make for a good life. Wellbeing includes material wellbeing, such as income and wealth; physical and psychological well-being, represented by health and happiness; and education and the ability to participate in civil society through democracy and the rule of law. Much of this book will focus on

two of these components, health and wealth; in this overview I also say something about happiness.

I start with an overview of wellbeing in the world as it is today, and how it has changed over the last thirty to fifty years. I present the basic facts with only minimal explanation; in later chapters I shall explore individual topics in more detail, ask how we got here, and where and how we should be going next.

Health and Wealth

Health is the obvious starting point for an enquiry into wellbeing. You need a life to have a *good* life, and poor health and disability among the living can severely limit the capability to enjoy an otherwise good life. So I begin with life itself.

A girl born in the United States today can expect to live for more than 80 years. This official estimate is in fact very conservative because it ignores any future reductions in mortality that might take place during her life; given past progress, it is implausible that progress will suddenly stop. Of course, it is hard to project future health improvements, but a reasonable guess would be that a white, middle-class girl born in affluent America today has a 50-50 chance of making it to 100.¹ This is a remarkable change from the situation of her great-grandmother, born in 1910, say, who had a life expectancy at birth of 54 years. Of all the girls born in the United States in 1910, 20 percent died before their fifth birthday, and only two out of every five thousand lived to celebrate their hundredth birthday. Even for her grandmother, born in 1940, life expectancy at birth was 66, and thirty-eight out of every thousand girls born in 1940 did not make it to their first birthday.

These historical differences pale in comparison with the differences between countries today. There are many places in the world whose health today is worse than it was in the United States in 1910. A quarter of all children born in Sierra Leone (or Angola or Swaziland or the

Democratic Republic of the Congo or Afghanistan) will not live to see their fifth birthday, and life expectancy at birth is only a little above 40. Women typically bear between five and seven children, and most mothers will suffer through the death of at least one of their children. In these countries, one out of every thousand births leads to the death of the mother, a risk that accumulates to one in a hundred for women who have ten children. Bad though these numbers are, they are much *better* than those a few decades ago: even in the worst places, where nothing else seems to go right, the chances of dying have been falling. In some of the countries with the worst outcomes, such as Swaziland, if children make it past age 5, they run the risk of HIV/AIDS, which has greatly increased the risk of dying in the young adult years—a time of life when very few usually die. But such horrors are not universal in tropical countries, or even in all poor countries. There are many countries, including at least one tropical country (Singapore), where a newborn has survival chances as good as or better than those in the United States. Even in China and India (which in 2005 contained between them more than a third of the world's people and almost half of the world's poorest people), newborns today can expect to live for 64 years (India) and 73 years (China.)

Later in this chapter, I say more about where these numbers come from, but it is worth emphasizing now that the poorer the country, the worse its health statistics tend to be. Even so, we have good information about the deaths of children—the fractions who die before age 1 or age 5—but much worse information about adult deaths—including maternal mortality rates, or how long a 15-year-old can expect to live.

Health is not just a matter of being alive, and living a long time, but of living in good health. Good health has many dimensions and is harder to measure than the mere fact of whether or not someone is alive, but here too there is evidence of improvement over time as well as of differences between rich and poor countries. People in rich countries report less pain and less disability than people in poor countries. Disability has been falling over time in rich countries. IQ

scores are rising over time. In most of the world, people are getting taller. Those who do not get enough to eat as children, or who live through childhood diseases, often do not grow as tall as their genes would have allowed under ideal conditions. Being shorter than they ought to have been may indicate early life misfortunes that compromise brain development and further restrict their opportunities in adult life. Europeans and Americans are taller than Africans, on average, and much taller than Chinese or Indians. Grown children are taller than their parents, and taller still than their grandparents. Global improvements in health and income, as well as global inequalities, can even be seen in people's bodies.

Differences in health are often mirrored by differences in material living standards or in poverty. Americans are much richer than they were in 1910 or in 1945, and the countries with the lowest life expectancies today have incomes that are a(n almost incredibly) tiny fraction of American incomes today. The (grotesquely misnamed) Democratic Republic of the Congo (DRC, and known as Zaire under the rule of Joseph Mobutu from 1965 to 1997) has a per capita national income that is about three-quarters of 1 percent of the per capita national income of the United States. More than half the population of the DRC lives on less than a dollar per person per day; the fractions for Sierra Leone and Swaziland are similar. Some of the worst places are not even documented because they are currently engulfed in conflict; Afghanistan is one example.

According to the U.S. Census Bureau, 14 percent of the American population was poor in 2009, but the poverty line in the United States is much higher, around \$15 a day. It is hard to imagine living in the United States on a dollar per person per day (although one calculation suggests that \$1.25 is possible if we exclude the cost of housing, health, and education),² yet this, or something close to it, is typical of what the very poorest people in the world survive on.

The link between life expectancy and poverty, although real enough, is far from exact. In China and India, with life expectancies of 73 and 64 years, respectively, many people live on less than a dollar a day—about a quarter of the population in India and a seventh of the rural population

in China. And although the Chinese economy will soon overtake the U.S. economy in total size, per capita incomes in China are only around 20 percent of American incomes; on average five Chinese share the income of one American. There are other, even poorer countries that do well in life expectancy. Bangladesh and Nepal, with life expectancies in the mid-60s, are examples; Vietnam is only a little better off but in 2005 had a life expectancy of 74.

There are also some rich countries that do much worse than their incomes warrant. A notable example is the United States, whose life expectancy is one of the lowest among the richest countries. Another case of a different kind is Equatorial Guinea, which in 2005 had a per capita income that was bloated by oil revenues, but a life expectancy of less than 50 years. Equatorial Guinea, once a Spanish colony in West Africa, is ruled by President Teodoro Obiang Nguema Mbasogo, who has a good claim to the highly contested title of Africa's worst dictator and whose family is the beneficiary of most of the country's revenue from exporting oil.

High life expectancy, good health, the absence of poverty, democracy, and the rule of law are among the features that we would include were we to design an ideal country. They allow people to live good lives and to pursue what is important to them. Yet, without asking people, we don't know *exactly* what they care about, how they might trade off health and income, or even the extent to which these things matter to them at all. People are sometimes capable of adapting to what might seem intolerable conditions, and perhaps they can extract some modicum of happiness or even live a good life in places where mortality and poverty are common—as it were, prospering in the valley of the shadow of death. Poor people may report that they are living good lives in the most difficult conditions, and rich people, who seem to have everything, may feel that their lives are deeply unsatisfactory.

In such cases, we might still choose to measure their wellbeing in terms of the opportunities that people have to lead a good life, rather than what they themselves make of their lives. That a poor man is happy and adaptable does not detract from his poverty any more than the

misery or greed of a billionaire detracts from his wealth. A focus on what Amartya Sen calls “capabilities” leads to an examination of freedom from deprivation in terms of the possibilities that are opened up by objective circumstances, rather than what people make of, or feel about, those circumstances.³ Yet feeling that one’s life is going well is good in and of itself, and it is better to be happy than to be sad. Those feelings contribute to a good life, and it is important to ask people about them, even if they are not given any special priority in the assessment of wellbeing. This is a different position from that proposed by some utilitarians, such as the economist Richard Layard,⁴ who argue that self-assessed happiness is the only thing that is important, that good circumstances are good only insofar as they promote happiness, and that bad circumstances are not bad if people are happy in spite of them. Even so, as we saw from Figure 1 and 2 in the Introduction, it turns out that people are not at all content with their lives in countries where life is nasty, brutish, and short, and that the inhabitants of the rich, long-lived countries are generally well aware of their good fortune.

Life Expectancy and Income in the World

To look at the general patterns—as well as to pick out the exceptions, which are often of great interest—we need to look at the world as a whole, mapping patterns of health, wealth, and happiness. One of the most useful ways of doing this was pioneered by the demographer Samuel Preston in 1975.⁵ Preston’s picture, updated to 2010, is redrawn in Figure 1; it shows life expectancy and income around the world.

The horizontal axis shows GDP per capita of each country while the vertical axis shows life expectancy at birth for men and women taken together. Each country is shown as a circle, and the areas of the circles are proportional to population size. The huge circles in the middle of the plot are China and India, while the considerably smaller but still large circle at the top right is the United States. The curve that runs from bottom left to top right illustrates the general relation

between life expectancy and national income, rising rapidly among the low-income countries and then flattening out among the rich, long-lived countries.

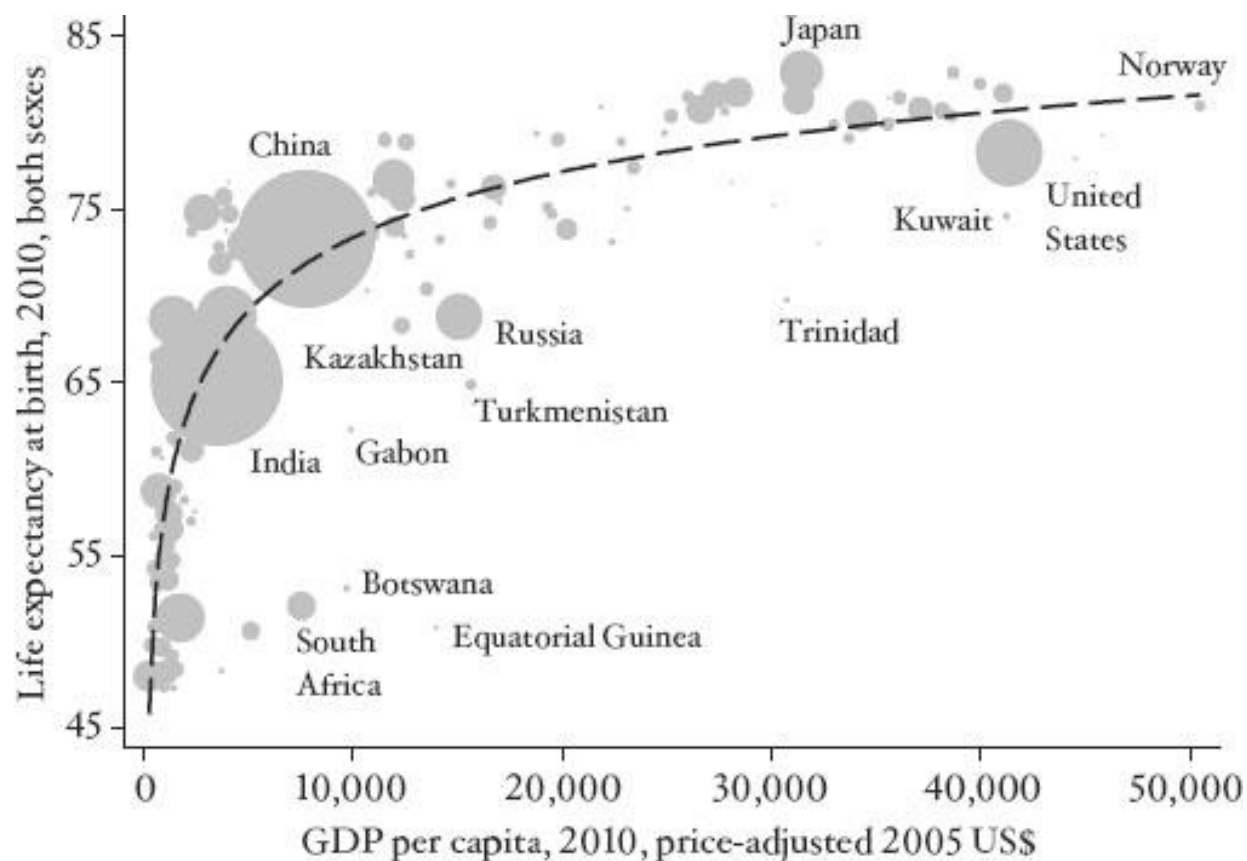


FIGURE 1 Life expectancy and GDP per capita in 2010.

GDP per capita is a measure of average income originating in each country and is measured here in a common unit across countries. The unit, the 2005 international dollar, is constructed so that, at least in principle, a dollar is worth the same in all countries and we are comparing like with like; an international dollar in Brazil or Tanzania buys the same as a dollar in the United States. GDP includes incomes that are not directly received by people or families, such as government tax receipts and the profits of firms and banks, as well as incomes that belong to foreigners. Generally only a fraction, albeit a substantial fraction, of GDP is available to households for their own

purchases. Other components of GDP benefit households directly (government expenditures on education, for example), or indirectly (investment for the future). GDP, which is gross *domestic* product, is different from GNP, gross *national* product. GNP includes, and GDP excludes, income owned by residents but generated abroad, and GNP excludes, and GDP includes, incomes generated domestically but owned by foreigners. The difference is usually small, but it is very important for some countries. Luxembourg, where many earners live in Belgium, France, or Germany, is an example of a country where GNP is much smaller than GDP. Another is the tiny Chinese peninsula of Macau, now the world's largest gambling casino. These two countries, which would appear beyond the right-hand boundary of the graph, are excluded, along with Qatar and the United Arab Emirates, oil-rich states that, along with Luxembourg and Macau, had the world's highest levels of GDP per head in 2010. GNP is a better measure of national income, but the data are more consistently available for GDP, which is why I use it here and in many places throughout the book.

An important feature of the graph is the “hinge point” near China where the curve begins to flatten. The hinge point marks the *epidemiological transition*. For countries to the left of the transition, infectious diseases are important causes of deaths, and many of the deaths are among children, so that in the poorest countries, about half of all deaths are of children under the age of 5. After the transition, as we move to the richer countries, child deaths become quite uncommon, and most deaths are of old people, who die not from infectious disease but from *chronic* diseases, the most important of which are heart disease (or more broadly, cardiovascular disease, including stroke) and cancer. Chronic diseases are becoming increasingly common causes of death in poor countries, too, but few people in rich countries die of infectious disease, except for small numbers of the elderly who die from pneumonia. One way in which the transition is sometimes summarized is to say that diseases move out of the bowels and chests of infants into the arteries of the elderly.

That life expectancy and incomes are positively related is important for thinking about the

worldwide distribution of wellbeing. Health and wealth are two of the most important components of wellbeing, and the graph shows that they generally (although not inevitably) go together. People who suffer deprivation in terms of material living standards—such as much of the population of sub-Saharan Africa—are generally also the people who suffer deprivation in terms of health; they get to live for fewer years, and they live with the misery of seeing many of their children die. At the other end of the curve, among the rich of the world, few parents ever experience the death of a child, and they get to enjoy their high standard of living for almost twice as many years as those in the poorest countries. Looking at the world in terms of health and income together forces us to see that the divisions are compounded, and that the dispersion of wellbeing is wider than it appears if we look only at health or at income. One crude and sometimes useful (although ethically unattractive) trick is to combine life expectancy and income by multiplying the two to give a measure of lifetime income. This is a poor measure of wellbeing (an extra year of life is valued by the income of the recipient, so a year of the life of a rich person is worth more than a year of the life of a poor person), but it illustrates the effects on the gaps between countries. In the DRC, for example, per capita income is estimated to be about three-quarters of 1 percent of that of the United States, and life expectancy is less than two-thirds of that of the United States, so that average lifetime income in the United States is more than two hundred times the average lifetime income in the DRC.

What the figure does *not* establish is that it is higher income that *causes* better health, or that it is poverty that causes what are often called the “diseases of poverty.” Nor does the graph rule it out, and indeed income *must* be important in some ways and at some times—an idea that will be explored extensively in the rest of this book. Income is important in places where improving health requires better nutrition—for which people need money—or cleaner water and better sanitation—for which governments need money. Among the rich countries, it is less obvious how money can tackle cancer or heart disease—though research and development are certainly

expensive—so that we have the beginnings of a simple account of the flattening of the curve as countries pass through the epidemiological transition. It is also possible that there is some upper limit to human life expectancy—perhaps surprisingly, the idea is hotly contested—so that, when life expectancy is as high as it is Japan, or even the United States, it gets harder and harder push it up further.

It is sometimes claimed that there is *no* relation between income and life expectancy among the better-off countries of the world.⁶ As was the case for the graphs of life evaluation versus GDP in the Introduction, it is useful to redraw Figure 1 using a log scale for income. Figure 2, which uses exactly the same data as Figure 1, gives a very different impression. To a first approximation, the slope of the line is the same on the right as on the left of the picture, although the relationship at the top is a little flatter—largely driven by the poor performance of the United States—and among the very richest countries the lack of a relationship is still apparent. But for much of the world, proportional increases in income are associated with the same increase in years of life, just as they are associated with the same increase in life satisfaction, as we saw in the Introduction. Of course, because the rich countries have much higher incomes, the same proportional increase in a rich country involves a much larger absolute increase than in a poor country, so that, as in Figure 1, the same amount of money comes with fewer additional years of life among the rich than among the poor. But even among the rich countries, higher incomes still come with more years of life. However, as Figure 2 shows, the ranking of countries by life expectancy is far from identical to the ranking by income.

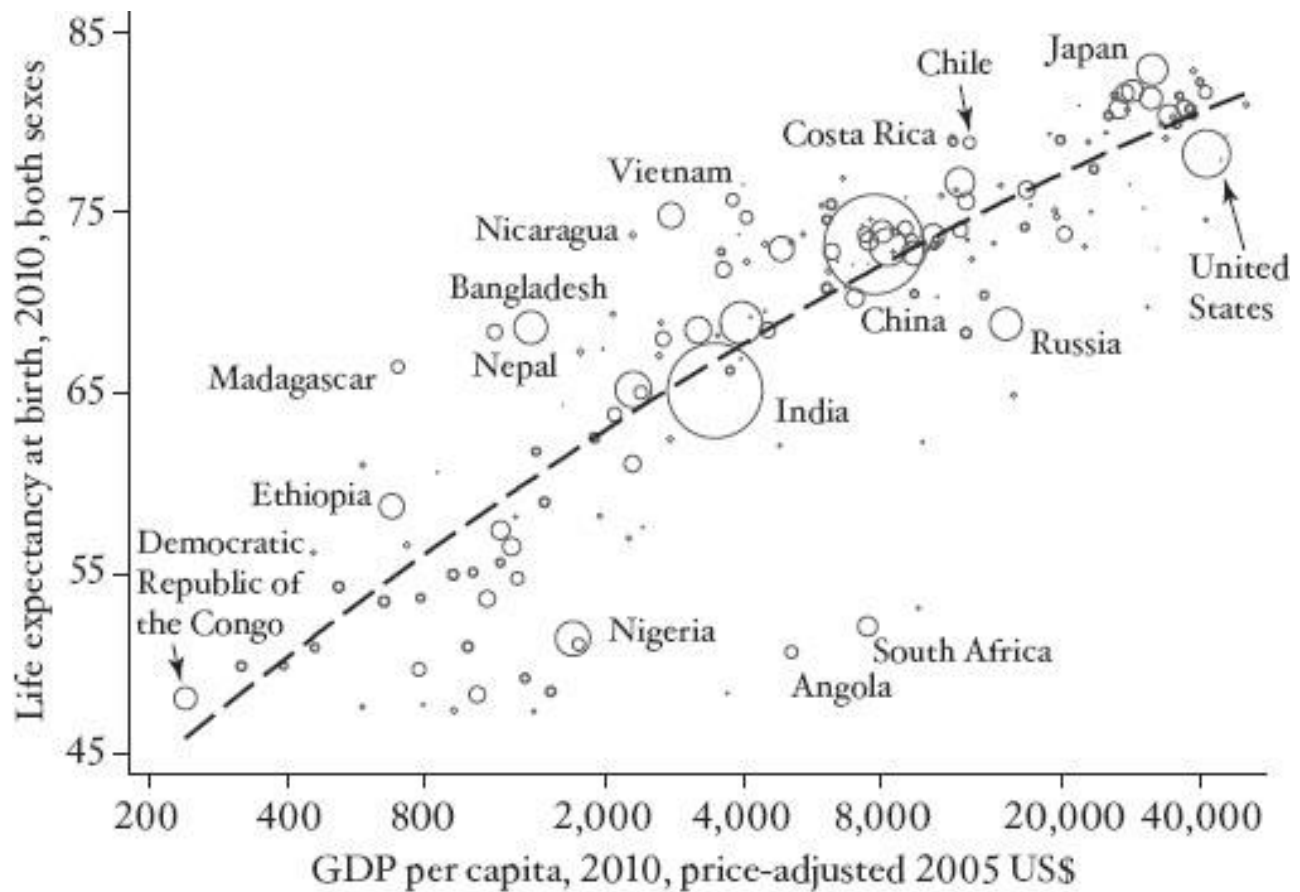


FIGURE 2 Life expectancy and GDP per capita in 2010 on a log scale.

The stories of the countries off the curve are as important as the stories of those that are on it. Among the countries that do much worse than might be expected given their levels of income, some have been affected by wars. Others—including Botswana and Swaziland (as well as other African countries not highlighted)—are suffering from the HIV/AIDS epidemic, which in several countries has taken back all or most of the gains in life expectancy achieved since World War II. For those countries, the disease has shifted them down and away from the curve. I have already discussed Equatorial Guinea, which is the most egregious of all. But the same factor—extreme inequality of income—is also partly responsible for South Africa’s position, which has been below

the curve for many years, long before the advent of HIV/AIDS. South Africa can be thought of—even after the demise of apartheid—as a small, rich country embedded in a much larger, poor country. Indeed, if we were to draw a line joining the United States to Nigeria in Figure 1, and then move 10 percent of the way from the United States to Nigeria—10 percent being the share of the white population in South Africa—we come close to South Africa’s position on the graph.

Russia is another of the large poor performers. It is a country where life expectancy decreased rapidly after the fall of communism, possibly in response to the chaos and disruption of the transition; consistent with this story, excess alcohol consumption was one of the precipitating factors, especially among men. What happened in Russia remains disputed, if only because mortality among men was increasing well before the change in political system.⁷ Whatever the truth, Russia and the countries of the former Soviet Union are places where both health and life evaluation are worse than might be expected given their incomes. They are also places where the transition from one economic system to another brought difficulties in the measurement of income, which may well be overstated in the figures. The transition in Russia, although inevitable in some form and probably beneficial in the long run, brought enormous costs in lost incomes and lost years of life. It does not rank with some of the other catastrophes of the postwar world—the AIDS epidemic or the Great Chinese Famine—but there was still enormous suffering and loss of wellbeing.

The United States is a poor performer relative to its income. Yet the United States spends a larger share of its national income on health care than any other country, so it provides a good illustration of the fact that there is no tight relation between income and health, and even less between health and expenditures on health care. Chile and Costa Rica have as good life expectancy as the United States, at about a quarter of per capita income and about 12 percent of per capita health expenditures. I shall return to U.S. health and health-care financing in Chapters 2 and 5.

Other countries do much better than might be expected from their incomes. Figure 2, with

the log scale, shows these more clearly than does Figure 1. Nepal, Bangladesh, Vietnam, China, Costa Rica, Chile, and Japan are important countries whose life expectancy is high relative to where we might expect it to be from the international curve. The poorest of these countries do well by managing to have unusually low infant (aged less than 1 year) and child (aged less than 5 years) mortality rates, while those at the top, especially Japan, have unusually low mortality among the middle-aged and elderly. I shall explore these exceptions in more detail later in the book, but the main point is that there is nothing predestined about the curve; poor countries can do better than would be expected given their resources, and rich countries can do worse. There are ways of ensuring good health at low incomes and ways of spending large sums of money to no purpose. War, epidemic disease, and extreme inequality also make health worse at any income level, although at least the first two are much more likely to occur among poor countries than rich countries.

Onward and Upward, with Catastrophic Interruptions

Figures 1 and 2 give a snapshot of the world in 2010. But the curve linking life expectancy and income has not stayed still. Figure 3 plots the data and two curves, one repeating the 2010 curve and one for 1960. Countries in 1960 are shown with lighter shading to distinguish them from countries in 2010. The areas of the circles are once again proportional to population, but within each year separately, so population change cannot be determined by comparing the size of the circle for a given country in 1960 with the size of the circle for that country in 2010.

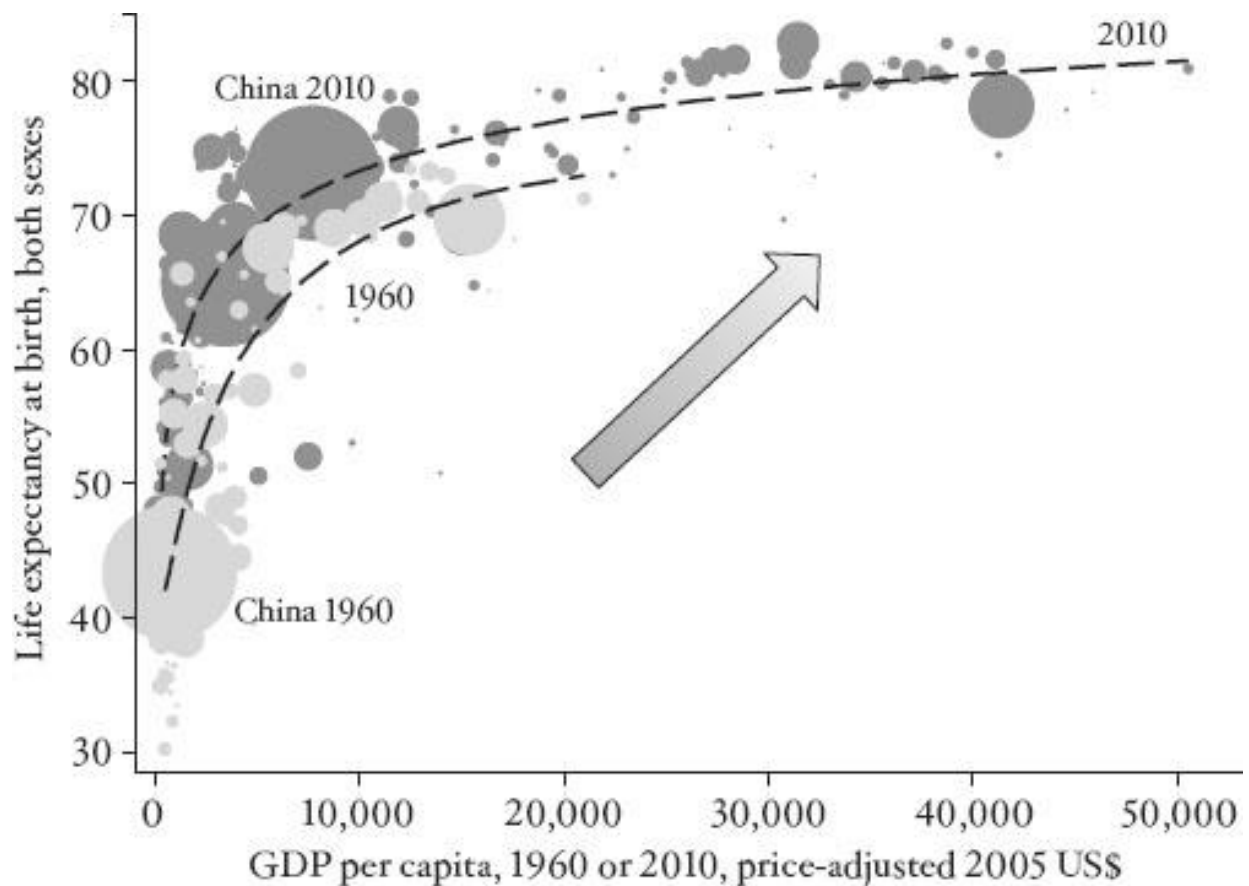


FIGURE 3 Longer lives, richer lives.

Almost all of the darker circles are above and to the right of the lighter circles; since 1960, nearly all countries have become richer and their residents longer lived. This is perhaps the most important fact about wellbeing in the world since World War II: that things are getting better, that both the health and income parts of wellbeing have improved over time. The economist and historian Robert Fogel, covering a longer span of history, has written about what he calls the escape from hunger and premature death.⁸ That Great Escape has continued apace throughout the world since World War II. Although a few countries have not escaped, and many more are only partway there, we should note and celebrate the successes. Many millions of people have escaped from a world of sickness and material deprivation. Amartya Sen writes of development as

freedom,⁹ and Figure 3 shows that the world is freer in 2010 than it was in 1960. And if we were to fill in the diagram with the (much less complete) information for 1930 or 1900, we would see that the expansion of freedom has been going on for a long time, starting around 250 years ago, gathering momentum, and involving more and more countries in the past half-century.

In spite of overall progress, there have been catastrophes. One of the worst in human history was China's "Great Leap Forward" in 1958–61, when deeply misguided industrialization and food-procurement policies led to the deaths of around thirty-five million people from starvation and prevented the births of perhaps forty million more. Weather conditions were not unusual in these years; the famine was entirely man-made.

Mao Zedong and his fellow leaders were determined to show the superiority of communism, to quickly overtake production levels in Russia and in Britain, and to establish Mao's leadership of the communist world. Outlandish production targets were set to match the food needs of rapidly industrializing cities and to earn foreign exchange through exports of food. Under the totalitarian system maintained by the Communist Party of China, rural communes competed to exaggerate their output, further inflating the already unattainable procurement quotas and leaving nothing for people to eat. At the same time, the Party caused chaos in the countryside by ordering that all private land be turned into communes, confiscating private property and even private cooking utensils, and making people eat in communal kitchens. Given the enormous increases in production that were confidently expected, peasant labor was diverted to public works projects and rural steel-making plants, most of which achieved nothing. Draconian restrictions on travel and communication prevented word from getting out, and the penalties for dissent were clear; three-quarters of a million people had been executed in 1950–51. (In any case, in these early years of the revolution, the Party was widely trusted.)

When Mao learned of the disasters (though probably not of their full scale), he doubled down on the policies, purging the messengers, labeling them "right-deviationists," and blaming

peasants for secretly hoarding food. To do otherwise and admit the error of the Great Leap Forward would have imperiled Mao's own leadership position, and he was prepared to sacrifice tens of millions of his countrymen to prevent that happening. If Mao had reversed course when the extent of the mass starvation first became clear to the leadership, the famine would have lasted one year, not three, and in any case there was more than enough grain in government stores to prevent everyone from starving.¹⁰

According to several accounts, life expectancy in China, which was nearly 50 in 1958, fell to below 30 in 1960; five years later, once Mao had stopped killing people, it had risen to nearly 55.¹¹ Nearly a third of those born during the Great Leap Forward did not survive it. We sometimes have a hard time identifying the benefits of policies, or even convincing ourselves that policy makes a difference. Yet the catastrophic effects of bad policies can be all too obvious, as the Great Leap Forward shows. Even in the absence of war or epidemic disease, bad policy within a totalitarian political system caused the deaths of tens of millions of people. Of course, bad policies happen all the time without causing millions to die. The problem in China was that the policy took so long to be reversed because of the totalitarian system and the lack of any mechanism to make Mao change course. The political system in China today is not so different from the system that Mao created; what is different is the flow of information. In spite of continuing state control, it is hard to believe that such a famine could happen today without the Chinese leadership, and the rest of the world, finding out very quickly. Whether the rest of the world would be able help any more today than it could then is far from clear.

The HIV/AIDS epidemic has been another great disaster. As we have already seen, it has raised mortality and dramatically decreased life expectancy in many countries in sub-Saharan Africa. The position of South Africa provides a graphic illustration. In Figures 1 and 2, we see South Africa far below the curve. If we go back to 1960, long before HIV/AIDS had any effect on mortality, it was in a very similar position—not because of disease, but because of the extreme

inequality between its white and black populations. If we were to run these curves as a movie and watch them change, decade by decade, we would see South Africa moving upwards, closer and closer to the curve as apartheid crumbled and racial differences in health narrowed. Or at least that was what happened up to 1990. After that, with increasing deaths from AIDS, the country tumbled back to its original position, falling back to where we see it in Figure 1.

In the past few years, antiretroviral drugs have begun to stanch the loss of life in Africa. The epidemic itself is another reminder that escapes can be temporary, and that great epidemics of infectious disease—HIV/AIDS now, cholera in the nineteenth century, and the Black Death in medieval times—are not safely confined to the past. Much attention, in both the scientific and the popular press, has been paid to current threats from “emerging” infectious diseases, particularly those, like HIV/AIDS, that crossed from animal reservoirs to humans. There are many such “zoonotic” diseases, some spectacularly and quickly lethal. Yet this is a lethality that makes it almost impossible for them to turn into large-scale epidemics; killing victims is good for neither the victims nor the bugs. HIV/AIDS, which is not easily transmitted and which kills very slowly, poses a much greater danger, and the pandemic that it caused should discourage us from believing that such diseases can be safely ignored in the future.

Turning away from catastrophes, we can see from Figure 3 not only that countries are getting richer and healthier, but that the curve linking life expectancy and income is itself moving up over time. The 2010 curve is above the 1960 curve, and if we were to go back through time, we would see that the 1960 curve is above the 1930 curve, which is above the 1900 curve, and so on. This upward movement was noted by Preston, who concluded that some systematic factor *other than income* must be responsible. If income were the most important thing—with other factors, like epidemic disease or country health policies, more or less without any pattern—then countries would move up or down (mostly up) the curve. But while countries did indeed move up the curve, that is not all that happened. Even with no change in income, life expectancy improved over time,

and did so the world over, at low and high income levels. Preston attributed this upward movement of the curve to improvements in scientific and medical knowledge, or at least to the greater practical implementation of existing scientific and medical knowledge. He thought of movements *along the curve* as the contribution of improved living standards to health, and movements *of the curve itself* as the contribution of new practical knowledge.¹² This division of the credit for increases in well-being between income and knowledge will occupy us throughout the book. I shall argue that it is knowledge that is the key, and that income—although important both in and of itself and as a component of wellbeing, and often as a facilitator of other aspects of wellbeing—is not the ultimate cause of wellbeing.

Global Poverty and Global Inequality

Material living standards are improving in most countries of the world. Yet there is nothing in logic that guarantees an automatic link between growth and reductions in global poverty; it might be that the poorest countries in the world are not growing at all—as was true in much of Africa in the 1980s and early 1990s—or it might be that, where there is growth, it has benefited only the already well-to-do within each country. Those who believe that globalization and economic growth are benefiting only the rich often make one or both of these arguments. Certainly, as we have already seen, there are almost unimaginable differences in average material living standards between countries, and the gaps between rich and poor within each country are hardly less wide. Are these inequalities getting wider with general economic progress? Is everyone benefiting or is it only the already rich who made the Great Escape, leaving the less fortunate behind?

One way to answer the question is to see whether initially poor countries have grown faster than initially rich countries, something that must happen if the gaps between them are to narrow. If it is progress in science and practical knowledge that makes economic growth possible, then we

might expect country living standards to move closer to one another, at least if the knowledge and techniques can easily be transferred from one country to another.

Start with Figure 4, which shows a more or less random scatter of dots. Each point in the figure is a country and shows its average per capita growth rate on the vertical axis against its initial GDP per person on the horizontal axis. The dark circles take 1960 as the starting point and look at growth from 1960 to 2010, while the lighter circles start from 1970 and show growth from 1970 to 2010. The lack of any patterns in the dots means that the poor countries did not grow faster than the rich countries, so that there was no catch-up, and no reduction in the inequality between countries. Nor did the rich countries grow faster than the poor countries. Overall, cross-country inequality did not change by very much. Nearly all of the growth rates are positive and lie above the broken line that indicates zero growth. There has been lots of growth in the world over the past half-century; only four countries had lower per capita incomes in 2010 than in 1960, and only fourteen had lower incomes than in 1970. As always, we should remember that some of the worst-performing countries (e.g., those at war) are excluded because there are no data for them, or because they did not exist in the earlier years. (The two worst performers in Figure 4 are the DRC and Liberia, both of which suffered wars.)

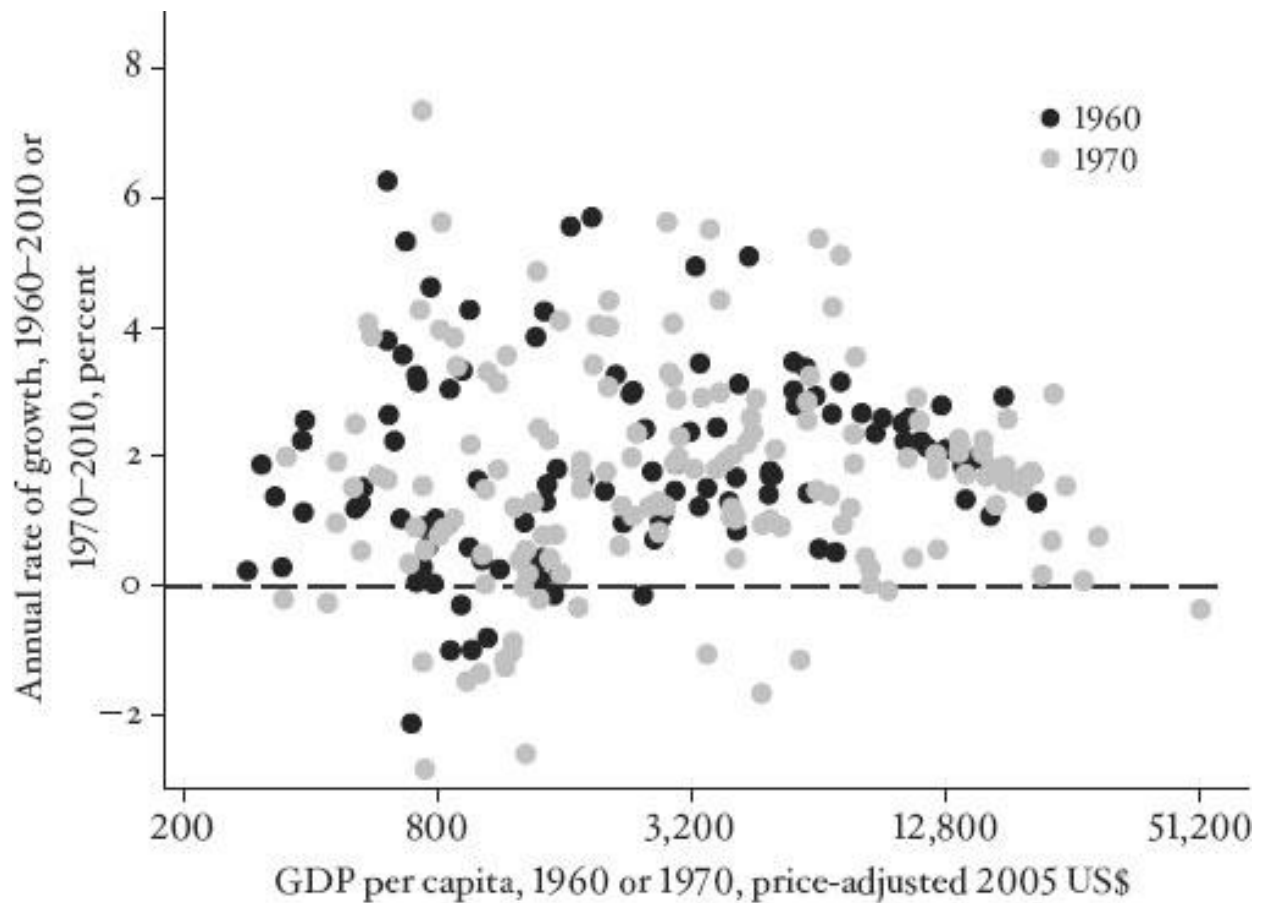


FIGURE 4 Growth, country by country.

There is a different and more positive way of looking at exactly the same data. Figure 5, which was first drawn in this way by economist Stanley Fischer, is identical to Figure 4, but each country is now plotted with a circle whose area is proportional to its population in the starting year.¹³ Looked at in this way, we get an immediate visual impression of a strong *negative* relation, with poorer countries growing faster. But we already know that the poorer countries did *not* grow more rapidly! The difference in perception comes from blowing up the dots for the biggest countries. The two largest countries in the world, China and India, have grown very rapidly over the past half-century. And because there are so many people in both, their growth has brought the average incomes of more than two billion people from the bottom of the world income

distribution—where they began—to somewhere much closer to the middle—where they are now. If each person in each country had the average income of the whole country, Figure 5 would show that the living standards of all the people in the world have drawn closer together, although there has been no narrowing of the average living standards of countries. Of course, it is not at all true that everyone shares the same income in each country; not only is there income inequality within countries but, as we shall see in Chapter 6, income inequality is widening in many (but not all) countries. Once within-country income inequality is taken into account, what is happening to income inequality over all the citizens of the world is much less clear, though a good case can be made that it is decreasing.

The rapid growth of China and India has not only enabled hundreds of millions of the world's citizens to make the Great Escape but made the world a more equal place. If we care about *people*, rather than *countries*, the optimistic picture in Figure 5, not the pessimistic picture in Figure 4, is the correct one.

The story of global poverty is also much affected by what has been happening in China and India. The World Bank regularly calculates the total number of people in the world who live in households whose daily income is less than a dollar a person. The latest version of these numbers up to 2008, as calculated by the World Bank, is shown in Figure 6.¹⁴ The total number of dollar-a-day poor people in the world fell by three-quarters of a billion between 1981 and 2008 in spite of an increase in the total population of poor countries of about two billion. As a result the fraction of the world's population that lives below a dollar a day has fallen from more than 40 percent to 14 percent. Although the poverty rate has fallen in other regions of the world, the fall in absolute numbers of poor has been driven in large part by the rapid growth of China, so that, at least until the past ten years, the absolute number of non-Chinese poor has continued to increase. (As we shall see in Chapter 6, there is a case to be made that Indian statistics are missing an increasing share of what people actually have, so that these numbers would understate how much

progress India has made in reducing poverty.) For Africa south of the Sahara, the Bank calculates that 37 percent of the population lived on less than a dollar a day in 2008, down from a peak of 49 percent in 1993; African economies have been growing in recent years, albeit from low levels. As always, African data should be treated with a good deal of caution. So in global poverty, too, there is general progress—not everywhere and not at all times, but a quarter-century of global growth has done much to reduce global poverty.

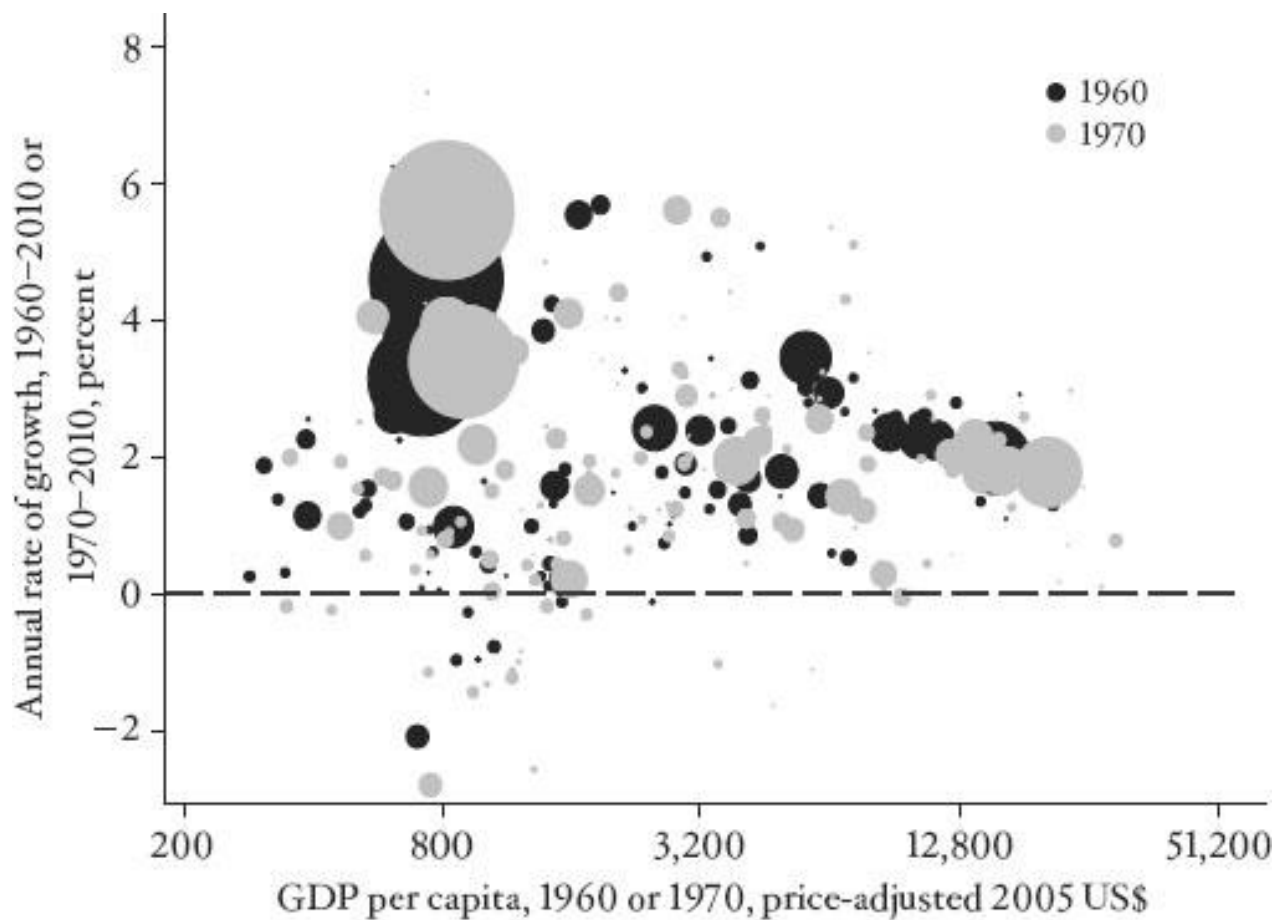


FIGURE 5 Growth weighted by country population.

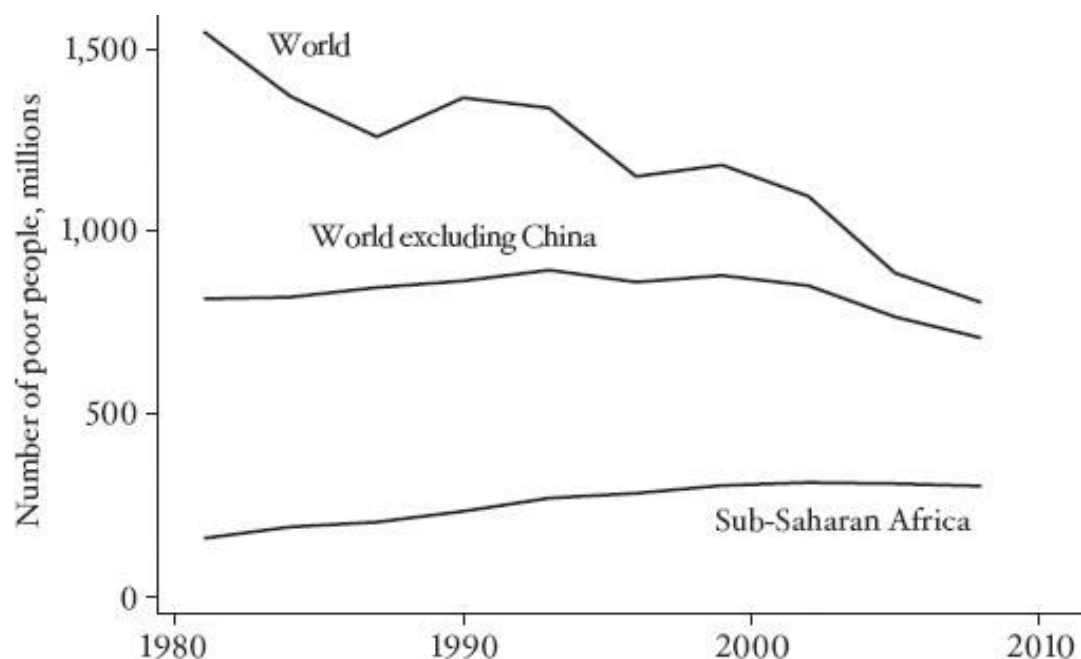


FIGURE 6 World poverty is falling.

How Do People See Their Own Lives?

Living a good life needs more than health and money, and the escape from deprivation that development brings should also involve better education and a better ability to participate in civic affairs. My main focus here is on health and income, but the general picture is very much the same: there has been great progress in recent decades, although much remains to be done. More children are regularly attending school, and more people are literate. And while there are many dictators, and many hundreds of millions of people live with (sometimes very severe) restrictions on their civic participation, there is more political freedom in the world now than was the case half a century ago. Life is getting better for most of the world, at least in terms of the opportunities that these circumstances allow.¹⁵ Yet it is always possible that people do not see their lives in terms of these measures, and that they are more prized by development experts or academic commentators

than by the people who are experiencing them. Or people may value different things that are not included in our lists. So there is a good deal to be said for asking people how *they* think their lives are going.

One way to do this is to use self-reported measures of wellbeing, like those described in Figures 1 and 2 of the Introduction. Economists, psychologists, and philosophers have recently become interested in these measures, and a number of national statistical offices are moving toward the routine collection of such data.¹⁶ These measures, which are often loosely referred to as measures of happiness, have many attractions: they come directly from the people whose wellbeing we are trying to assess, they measure actual achieved outcomes, and they might possibly include the effects of factors that are important for wellbeing, but which we do not know about, or which we know about but cannot measure.

Yet many writers, both economists and philosophers, have reservations about the validity and usefulness of self-reported measures of wellbeing. We do not always know what people are thinking when they answer these questions, and there are doubts about whether the questions are interpreted the same way by different people or by different nations. Translation of questions is sometimes difficult even when a direct translation exists; Americans use “happy” more freely and more frequently than the French use “heureux,” and East Asians seem especially reluctant to say that they are happy.¹⁷ In the United States, the pursuit of happiness is one of the unalienable rights enumerated in the Declaration of Independence, yet in the Calvinist Scottish village in which I grew up, such a pursuit would have been seen as indicating a serious weakness of character.

Of greater concern still is adaptation; people living in desperate circumstances can come to believe that this is the best that life can offer and report that they are happy. Others, who live in the lap of luxury, have become so used to wealth that they can become dissatisfied over the absence of trivial luxuries.¹⁸ A full and happy life can sometimes involve pain and loss; the philosopher Martha Nussbaum writes about the “happy warrior” going into battle, expecting to meet nothing

but pain and possible death, who nevertheless feels that he is leading a good and worthy life.¹⁹

These reservations do not mean that we should ignore what people say about their lives, only that we should be alert to the potential problems and not turn off our skepticism.

If people always adapt so as to be satisfied with what they have, the average answer should not vary much across countries; most of the rich countries of the world have been rich for a long time, and most of the poor countries have been poor for a long time, so that people have had plenty of time to get used to their circumstances. But the figures in the Introduction show that this is not what happens.

The life evaluation score for Denmark (perennially the best performer in these comparisons) is 7.97 (on a “ladder” scale with “steps” marked from 0 to 10), followed by the other Nordic countries—Finland 7.67, Norway 7.63, and Sweden 7.51—with the United States only a little behind them at 7.28. Togo (a long-standing dictatorship) is 2.81, Sierra Leone (after years of civil war) is 3.00, and Zimbabwe (yet another long-standing dictatorship) is 3.17. Burundi at 3.56, Benin at 3.67, and Afghanistan at 3.72 come next on the roll of misery. The philosophical doubts about these measures are real enough, but when it comes to assessing deprivation and identifying the countries where people do or do not flourish, the life evaluation measures are in close accord with measures of income, health, or political freedom. The rich, developed, and democratic countries of Europe, America, and the European offshoots are better places to live than the poorest countries of sub-Saharan Africa, Asia, and Latin America, and we get the same result from direct questions about how life is going as we do from looking at income or longevity.

It would be good to look at answers to the life evaluation question over the past half-century, to go back and compare what has happened since 1960, just as I did for the relationship between income and health. But Gallup’s World Poll began only in 2006, and although there are scattered data for a few countries from earlier years, we know little about the reliability of these numbers, or even how the respondents were selected. So we currently cannot

say whether or not the growth in the world over the past half-century has brought an increase in life evaluation.

Even so, that the residents of richer countries systematically value their lives more highly than those of poorer countries creates a strong presumption that growth is good for the way that people feel about their lives. The most obvious difference between Denmark and the United States on the one hand, and Sierra Leone, Togo, and Zimbabwe on the other, is that one group is rich and the other poor; that difference is the result of 250 years of growth in the rich countries versus none in the poor countries. There are also enormous differences in life expectancy, as we have already seen, but life expectancy too has increased along with economic growth in the past half-century. So it would be strange indeed if the average life evaluation in China, Germany, Japan, or the United States in 2008 would not be higher than those countries' average life evaluations in 1960. But this seemingly uncontroversial conclusion has been mired in debate.

In 1974, the economist and historian Richard Easterlin, a pioneer in measuring self-reported wellbeing, argued that economic growth in Japan had not made people's lives better by their own reports, and in subsequent work he extended the finding to a number of countries, including the United States.²⁰ His claim, then and now, is that economic growth does not improve the human lot. Easterlin is unusual among economists in arguing that growth is worthless in and of itself. (He does not question the health and other benefits that have come with—although not necessarily *because of*—economic growth.) His position resonates with many psychologists, religious leaders, and others who deny the materialist basis of wellbeing, except perhaps among the most deprived. The economists Betsey Stevenson and Justin Wolfers have challenged these beliefs and argue that, with properly comparable data, economic growth within countries improves life evaluation in exactly the way that we would expect from the differences in life evaluation between rich and poor countries.²¹

The effects of economic growth on a country's life evaluation are much harder to see than

are the effects of differences between rich and poor countries. Even fifty years of economic growth does not move a country very far compared with international differences that are the result of *centuries* of differences in growth rates. If a country were to maintain a constant rate of growth of 2 percent a year for half a century (about the average in Figure 4), its per capita national income would be 2.7 times larger at the end of the period. This is a substantial increase, but it is about the same as the difference between India and Thailand today. Given that countries do not lie exactly on the line linking life evaluation and income, it would not be surprising if such periods of economic growth were accompanied by increases in life evaluation that are small, hard to detect, or even perverse. Indeed, as Figure 1 in the Introduction shows, China, whose per capita income in 2008 was twice that of India's, has a substantially lower life evaluation score.

Just as there are countries whose health is better or worse than can be expected from looking at their incomes, there are countries whose residents rate their lives more or less highly than can be expected from their incomes. We have already seen that the Scandinavian countries are wellbeing superstars, but they are also very rich countries, and their life evaluation scores are not much higher than would be predicted from their national incomes. Another frequent finding is that Latin American countries often do very well. Several of the East Asian countries do relatively badly, including China, Hong Kong, Japan, and Korea. We do not know whether these continental differences come from genuine differences in some objective aspect of wellbeing, from national differences in disposition, or from national differences in the way people respond to the ladder question. One frequent finding is an exceptionally low level of wellbeing for Russia, the countries of the former Soviet Union, and the countries of Eastern Europe that were once part of the communist bloc. It is the elderly who are particularly unhappy with their lives in these Eastern European and former Soviet countries.²² Young people there have opportunities that were not available to previous generations, including the chance to travel, to study abroad, and to find a place for their talents in the global economy. At the same time, their grandparents have seen the

collapse of the world that they knew and that gave meaning to their lives, and in some cases have also experienced disruptions to their pensions and health-care systems.

Emotional Wellbeing

Life evaluation measures are often described as measures of happiness, even though, as in the ladder question, there may be no mention of happiness in the question. There is now good evidence that life evaluation measures, which ask people to think about how their lives are going, pick up different aspects of experience and give different results from questions about feelings or experienced emotions. It is possible to be unhappy or worried, or to feel stress, even at times when you think your life is generally going well. Indeed, sadness, pain, and stress might be inevitable during some of the experiences one must go through in order to build a good life. Army boot camp, graduate studies in economics, medical school, or dealing with the death of a parent are examples of unpleasant experiences that are nevertheless an essential part of life; young people go on dates that sometimes result in dreadful experiences, yet they are a necessary part of emotional learning. These emotional experiences and others are important contributors to current wellbeing in their own right. Yet feeling happy is better than feeling sad, and stress, worry, and anger reduce wellbeing at the time they are being experienced, even if they sometimes have payoffs in the future.

Just as we can ask people to evaluate their own lives, we can also ask them about their emotional experiences. The Gallup World Poll, in addition to its life evaluation question, asks people about the emotions and feelings that they experienced the day before the survey, asking about worry, stress, sadness, depression, happiness, anger, and pain. It turns out that national average responses to these questions are quite different from national averages of life evaluation.

The global map of happiness is shown in Figure 7, which plots national income against the fraction of the population who report that they experienced happiness during a lot of the day

yesterday. This map is quite different from the map of life evaluation; most notably, there is a much weaker relationship with national income. Although it is true that some of the very poorest countries—such as Burkina Faso, Burundi, Madagascar, and Togo—report very little happiness, there is little systematic difference in happiness between rich and poor in countries other than the very poorest. Denmark, where people think that their lives are going extremely well, is not so good a place for experiencing happiness. Nor is Italy, and indeed a larger fraction of Bangladeshis, Kenyans, Nepalese, and Pakistanis experience a great deal of happiness than do Danes or Italians.

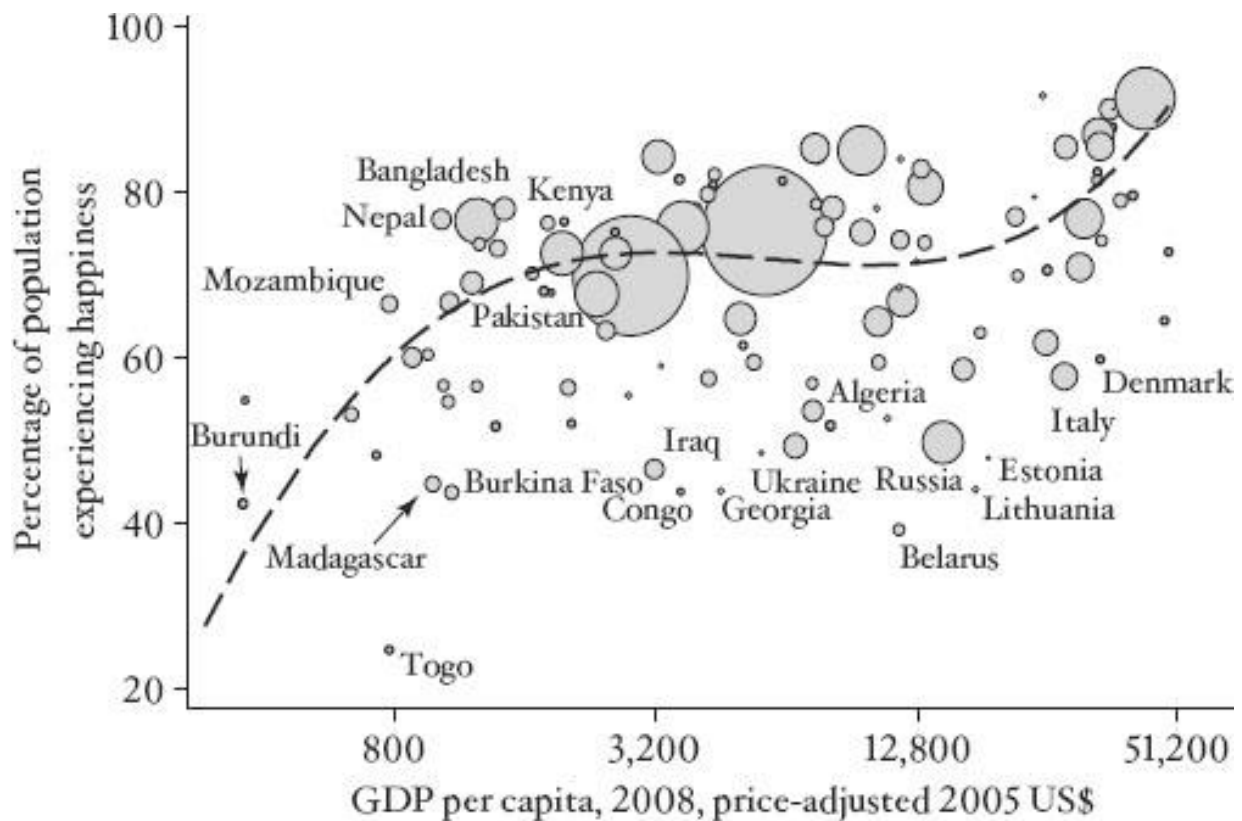


FIGURE 7 Happiness around the world.

The limited relationship between income and experienced happiness holds true within the United States too. Poverty generates misery, but beyond a certain point (about \$70,000 a year), additional money does nothing to improve happiness, even though those with more money report

that they have better lives.²³ For happiness money matters only up to a point. This is useful to know, if only because the experience of happiness is a positive one that makes life better. But it makes happiness a poor measure of overall wellbeing, because there are many places in the world where people manage to find happiness even in the midst of poor health and material poverty; life evaluation measures are much better measures of overall wellbeing. The cases of Denmark and Italy are good examples.

The happiness map shows that the United States, where being happy is something of a civic responsibility, is third from the top, bettered only by Ireland and New Zealand. Russia and its former satellites are among the unhappiest countries in the world. Yet most people in the world are happy; nearly three-quarters of the world's population report that they experienced happiness during a lot of the day yesterday.

Other measures of emotional experience provide different pictures yet again. In 2008, 19 percent of the world's population experienced anger during a lot of the day before the survey, 30 percent experienced stress, 30 percent experienced worry, and 23 percent experienced pain. There is more pain in poorer countries, though there is lots of variation at any given income level. But the national averages of worry, stress, and anger are not at all related to national income, although they too vary a great deal from one country to another. For example, three-quarters of Filipinos report a great deal of stress, followed by the citizens of Hong Kong, Lebanon, Syria, and the United States, where 44 percent report that they experienced stress during a lot of the previous day. National income seems to do little to relieve these negative emotions.

Life evaluation and happiness (or other emotions) paint different pictures of the world. Which is correct? This is a sensible question only if we expect such measures to yield a single measure of overall wellbeing, something that is a goal of much of the happiness literature. But this is not the right way to think about wellbeing. It is good to be happy, it is not good to be worried or to be angry, and it is good to think that your life is going well. But these feelings are not the same

thing, and all are consistent with good or bad outcomes in other aspects of wellbeing, such as income and physical and mental health. There is no magic question that provides a touchstone for judging wellbeing. Even if people carried a personal “hedonimeter,” which, like a wristwatch, would register happiness at every moment, there is no reason to suppose that the reading on the hedonimeter would be useful for assessing the goodness in their lives. Human well-being has many different aspects, often related but not the same, and if we are to measure the wellbeing of the world, we must recognize and do justice to that richness.

The historian Keith Thomas writes about changes in the ways that people sought personal fulfillment in England, and how, by the eighteenth century, the pursuit of wealth came to be seen as a legitimate and ethical route to happiness.²⁴ Adam Smith’s *Wealth of Nations* crystallized the long-evolving idea that the pursuit of wealth not only was a respectable activity for individuals but also brought benefits to society as a whole. Smith’s metaphor of the “invisible hand” has become part of our understanding of how capitalism works. Yet, as Thomas notes, Smith was skeptical about the personal benefits of wealth. Indeed, in his *Theory of Moral Sentiments*, Smith described the idea that wealth would make as happy as a deception, albeit a useful one “which rouses and keeps in perpetual motion the industry of mankind.” He was also skeptical about the extent of inequality, arguing that the rich, by employing others “in the gratification of their own vain and insatiable demands,” brought about an approximately equal distribution of the “necessaries of life.” As for the rich man, his great possessions “keep off the summer shower, not the winter storm, but leave him always as much, and sometimes more exposed than before, to anxiety, to fear, and to sorrow; to diseases, to danger, and to death.”²⁵

Smith wrote as the Great Divergence was about to begin, and in an age in which infectious disease threatened poor and rich alike. As we shall see in the next chapter, life expectancy was no higher for English aristocrats than for the common people. Even today, as we have just seen, the emotional lives of the poor are not very different from those of the rich, though they are much less

satisfied with their lives; riches are no protection against anxiety, fear, and sorrow, and they are not required to experience the happiness and enjoyment of everyday life. But the world has changed in the past 250 years. There is no reasonable interpretation of the evidence in which even the “necessaries of life” are equally distributed around the world, nor is it likely that they were so in Britain in Smith’s day. And riches today provide powerful protection against the dangers of disease and of death. As the world as a whole has become richer and has learned more, especially over the past sixty years, these protections have been extended to more and more of its population.

Income and health have improved almost everywhere since World War II. There is not a single country in the world where infant or child mortality today is not lower than it was in 1950.²⁶ Economic growth has propelled millions of people out of awful destitution, particularly in China and in India. Yet there have been terrible reversals. The Chinese famine, the HIV/AIDS pandemic, the collapse of longevity in the former Soviet Union, and scores of wars, massacres, and famines remind us that the curses of disease, war, and bad politics are not monsters safely confined to history. It would be rash indeed to suppose otherwise: as in the movie, the Great Escape might not bring permanent freedom but only a temporary reprieve from the evil, darkness, and disorder that surround us.